

Compiled Wrong Answers — All Models

Every non-fully-correct answer across all 13 models, grouped into four categories. Totals: 491 valid-but-wrong · 168 invalid · 187 row/side over-specifications · 253 correct-but-reasoning-blocked · 1099 rows overall.

gemma3-12b (google/gemma-3-12b-it · lenient 98.6% / strict 97.2%)

1. Valid but wrong 4 — answered a real label from the list, but not the correct one

#	Referring expression	Correct answer	Model said
#42	circle-row white shaft	GEAR_ROD_ROW1	GEAR_ROW1_LEFT
#58	the stand immediately to the left of the crank handle	STAND_ROW1_RIGHT	STAND_ROW1_LEFT
#141	larger gear of Row 2	GEAR_ROW2_RIGHT	GEAR_ROW2_LEFT
#271	a small silver screw with a recessed Phillips head	SCREW_ROW4	SCREW_ROW1

2. Invalid 0 — could not be matched to any of the 26 allowed labels

None.

3. Row/side over-specification 4 — answered a specific identifier (e.g. BEARING_ROW2_LEFT) instead of the grouped label (BEARING/PIN/SCREW_ROW{n})

#	Referring expression	Correct answer	Raw model output
#148	a screw at the base of a Row 2 gear stand	SCREW_ROW2	SCREW_ROW2_LEFT
#272	a screw at the base of a Row 4 gear stand	SCREW_ROW4	SCREW_ROW4_LEFT
#274	the silver Phillips screw for the blue Row 4 stand	SCREW_ROW4	SCREW_ROW4_RIGHT
#276	the small silver screw located at the foot of a blue support bracket	SCREW_ROW4	SCREW_ROW4_LEFT

4. Correct, but blocked by reasoning chain 0 — the model identified the right part, but the raw output still contains a <think> block or other wrapping, so strict grading (which compares the full raw text) can never match

None.

qwen3-vl-8b (Qwen/Qwen3-VL-8B-Instruct · lenient 98.6% / strict 94.1%)

1. Valid but wrong 4 — answered a real label from the list, but not the correct one

#	Referring expression	Correct answer	Model said
#58	the stand immediately to the left of the crank handle	STAND_ROW1_RIGHT	STAND_ROW1_LEFT
#208	larger gear of Row 3	GEAR_ROW3_RIGHT	GEAR_ROW3_LEFT
#209	green larger gear	GEAR_ROW3_RIGHT	GEAR_ROW3_LEFT
#211	larger green gear in the hexagon row	GEAR_ROW3_RIGHT	GEAR_ROW3_LEFT

2. Invalid 0 — could not be matched to any of the 26 allowed labels

None.

3. Row/side over-specification 13 — answered a specific identifier (e.g. BEARING_ROW2_LEFT) instead of the grouped label (BEARING/PIN/SCREW_ROW{n})

#	Referring expression	Correct answer	Raw model output
#81	a screw at the base of a Row 1 gear stand	SCREW_ROW1	SCREW_ROW1_RIGHT
#83	the silver H5 hex-socket screw for the white Row 1 stand	SCREW_ROW1	SCREW_ROW1_RIGHT
#84	the Row 1 fastener tightened with a 5 mm hex key	SCREW_ROW1	SCREW_ROW1_RIGHT
#85	the small silver screw located at the foot of a white support bracket	SCREW_ROW1	SCREW_ROW1_LEFT
#148	a screw at the base of a Row 2 gear stand	SCREW_ROW2	SCREW_ROW2_RIGHT
#151	the Row 2 fastener tightened with a T25 driver	SCREW_ROW2	SCREW_ROW2_RIGHT
#152	the small black screw located at the foot of a red support bracket	SCREW_ROW2	SCREW_ROW2_RIGHT
#215	a screw at the base of a Row 3 gear stand	SCREW_ROW3	SCREW_ROW3_RIGHT
#217	the black H3 hex-socket screw for the green Row 3 stand	SCREW_ROW3	SCREW_ROW3_RIGHT
#218	the Row 3 fastener tightened with a 3 mm hex key	SCREW_ROW3	SCREW_ROW3_RIGHT
#219	the small black screw located at the foot of a green support bracket	SCREW_ROW3	SCREW_ROW3_RIGHT
#271	a small silver screw with a recessed Phillips head	SCREW_ROW4	SCREW_ROW4_RIGHT
#276	the small silver screw located at the foot of a blue support bracket	SCREW_ROW4	SCREW_ROW4_LEFT

4. Correct, but blocked by reasoning chain 0 — the model identified the right part, but the raw output still contains a <think> block or other wrapping, so strict grading (which compares the full raw text) can never match

None.

qwen25-7b (Qwen/Qwen2.5-7B-Instruct · lenient 95.8% / strict 89.2%)

1. Valid but wrong 12 — answered a real label from the list, but not the correct one

#	Referring expression	Correct answer	Model said
#54	left bracket of the white circle row	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#58	the stand immediately to the left of the crank handle	STAND_ROW1_RIGHT	STAND_ROW1_LEFT
#65	white right gear stand in the circle row	STAND_ROW1_RIGHT	STAND_ROW2_RIGHT
#66	right bracket of the white circle row	STAND_ROW1_RIGHT	STAND_ROW2_RIGHT
#133	smaller red gear in the triangle row	GEAR_ROW2_LEFT	GEAR_ROW2_RIGHT
#178	left bracket of the green hexagon row	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#208	larger gear of Row 3	GEAR_ROW3_RIGHT	GEAR_ROW3_LEFT
#209	green larger gear	GEAR_ROW3_RIGHT	GEAR_ROW3_LEFT
#211	larger green gear in the hexagon row	GEAR_ROW3_RIGHT	GEAR_ROW3_LEFT
#268	blue gear in the square row	GEAR_ROW4_LEFT	GEAR_ROW3_RIGHT
#271	a small silver screw with a recessed Phillips head	SCREW_ROW4	SCREW_ROW1
#278	Phillips screw	SCREW_ROW4	SCREW_ROW1

2. Invalid 0 — could not be matched to any of the 26 allowed labels

None.

3. Row/side over-specification 19 — answered a specific identifier (e.g. BEARING_ROW2_LEFT) instead of the grouped label (BEARING/PIN/SCREW_ROW{n})

#	Referring expression	Correct answer	Raw model output
#16	the black bearing that supports a rotating gear shaft inside a stand	BEARING	BEARING_ROW2_LEFT
#18	black bearing	BEARING	BEARING_ROW2_LEFT
#79	a Row 1 H5 hex-socket mounting screw	SCREW_ROW1	SCREW_ROW1_LEFT
#81	a screw at the base of a Row 1 gear stand	SCREW_ROW1	SCREW_ROW1_LEFT
#82	a fastener used to secure a Row 1 bearing stand to the baseboard	SCREW_ROW1	SCREW_ROW1_RIGHT
#83	the silver H5 hex-socket screw for the white Row 1 stand	SCREW_ROW1	SCREW_ROW1_LEFT
#84	the Row 1 fastener tightened with a 5 mm hex key	SCREW_ROW1	SCREW_ROW1_RIGHT
#213	a Row 3 H3 hex-socket mounting screw	SCREW_ROW3	SCREW_ROW3_LEFT
#215	a screw at the base of a Row 3 gear stand	SCREW_ROW3	SCREW_ROW3_LEFT
#216	a fastener used to secure a Row 3 bearing stand to the baseboard	SCREW_ROW3	SCREW_ROW3_RIGHT
#217	the black H3 hex-socket screw for the green Row 3 stand	SCREW_ROW3	SCREW_ROW3_LEFT

#	Referring expression	Correct answer	Raw model output
#218	the Row 3 fastener tightened with a 3 mm hex key	SCREW_ROW3	SCREW_ROW3_RIGHT
#219	the small black screw located at the foot of a green support bracket	SCREW_ROW3	SCREW_ROW3_RIGHT
#270	a Row 4 Phillips mounting screw	SCREW_ROW4	SCREW_ROW4_RIGHT
#272	a screw at the base of a Row 4 gear stand	SCREW_ROW4	SCREW_ROW4_LEFT
#273	a fastener used to secure a Row 4 bearing stand to the baseboard	SCREW_ROW4	SCREW_ROW4_RIGHT
#274	the silver Phillips screw for the blue Row 4 stand	SCREW_ROW4	SCREW_ROW4_RIGHT
#275	the Row 4 fastener tightened with a Phillips screwdriver	SCREW_ROW4	SCREW_ROW4_RIGHT
#276	the small silver screw located at the foot of a blue support bracket	SCREW_ROW4	SCREW_ROW4_LEFT

4. Correct, but blocked by reasoning chain 0 — the model identified the right part, but the raw output still contains a <think> block or other wrapping, so strict grading (which compares the full raw text) can never match

None.

gemma4-e4b (google/gemma-4-E4B-it · lenient 93.8% / strict 92.7%)

1. Valid but wrong 16 — answered a real label from the list, but not the correct one

#	Referring expression	Correct answer	Model said
#53	white left gear stand in the circle row	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#54	left bracket of the white circle row	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#58	the stand immediately to the left of the crank handle	STAND_ROW1_RIGHT	STAND_ROW1_LEFT
#102	the left bearing support bracket in the second row from Row 1	STAND_ROW2_LEFT	BEARING
#108	Row 2 left bracket	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#111	left bracket of the red triangle row	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#114	the right bearing support bracket in the second row from Row 1	STAND_ROW2_RIGHT	BEARING
#120	Row 2 right bracket	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#123	right bracket of the red triangle row	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#178	left bracket of the green hexagon row	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#187	Row 3 right bracket	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#190	right bracket of the green hexagon row	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#233	square-row blue shaft	GEAR_ROD_ROW4	GEAR_ROW4_LEFT
#236	the left bearing support bracket in Row 4	STAND_ROW4_LEFT	BEARING
#245	left bracket of the blue square row	STAND_ROW4_LEFT	GEAR_ROW4_LEFT
#271	a small silver screw with a recessed Phillips head	SCREW_ROW4	SCREW_ROW1

2. Invalid 2 — could not be matched to any of the 26 allowed labels

#	Referring expression	Correct answer	Raw model output
#66	right bracket of the white circle row	STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT
#257	right bracket of the blue square row	STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT

3. Row/side over-specification 3 — answered a specific identifier (e.g. BEARING_ROW2_LEFT) instead of the grouped label (BEARING/PIN/SCREW_ROW{n})

#	Referring expression	Correct answer	Raw model output
#274	the silver Phillips screw for the blue Row 4 stand	SCREW_ROW4	SCREW_ROW4_RIGHT
#275	the Row 4 fastener tightened with a Phillips screwdriver	SCREW_ROW4	SCREW_ROW4_RIGHT
#276	the small silver screw located at the foot of a blue support bracket	SCREW_ROW4	SCREW_ROW4_LEFT

4. Correct, but blocked by reasoning chain 0 — the model identified the right part, but the raw output still contains a <think> block or other wrapping, so strict grading (which compares the full raw text) can never match
None.

qwen25-vl-7b (Qwen/Qwen2.5-VL-7B-Instruct · lenient 89.9% / strict 79.5%)

1. Valid but wrong 28 — answered a real label from the list, but not the correct one

#	Referring expression	Correct answer	Model said
#23	the handle immediately to the right of the right white gear stand	CRANK_HANDLE_ROW1	PIN
#30	right-side crank	CRANK_HANDLE_ROW1	PIN
#49	the upright white bearing bracket on the left, beside the large left gear in the circle-marked row	STAND_ROW1_LEFT	BEARING
#51	Row 1 left bracket	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#54	left bracket of the white circle row	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#58	the stand immediately to the left of the crank handle	STAND_ROW1_RIGHT	STAND_ROW1_LEFT
#66	right bracket of the white circle row	STAND_ROW1_RIGHT	GEAR_ROW1_LEFT
#106	the upright red bearing bracket on the left, beside the small left gear in the triangle-marked row	STAND_ROW2_LEFT	BEARING
#108	Row 2 left bracket	STAND_ROW2_LEFT	BEARING
#111	left bracket of the red triangle row	STAND_ROW2_LEFT	BEARING
#118	the upright red bearing bracket on the right, beside the medium right gear in the triangle-marked row	STAND_ROW2_RIGHT	BEARING
#123	right bracket of the red triangle row	STAND_ROW2_RIGHT	BEARING
#125	the small red spur wheel	GEAR_ROW2_LEFT	GEAR_ROW2_RIGHT
#133	smaller red gear in the triangle row	GEAR_ROW2_LEFT	GEAR_ROW2_RIGHT
#166	hexagon-row green shaft	GEAR_ROD_ROW3	GEAR_ROW3_LEFT
#173	the upright green bearing bracket on the left, beside the small left gear in the hexagon-marked row	STAND_ROW3_LEFT	BEARING
#175	Row 3 left bracket	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#178	left bracket of the green hexagon row	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#185	the upright green bearing bracket on the right, beside the medium right gear in the hexagon-marked row	STAND_ROW3_RIGHT	BEARING
#190	right bracket of the green hexagon row	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#208	larger gear of Row 3	GEAR_ROW3_RIGHT	GEAR_ROW3_LEFT
#209	green larger gear	GEAR_ROW3_RIGHT	GEAR_ROW3_LEFT
#211	larger green gear in the hexagon row	GEAR_ROW3_RIGHT	GEAR_ROW3_LEFT
#240	the upright blue bearing bracket on the left, beside the large left gear in the square-marked row	STAND_ROW4_LEFT	BEARING
#245	left bracket of the blue square row	STAND_ROW4_LEFT	GEAR_ROW4_LEFT
#252	the upright blue bearing bracket on the right, beside the crank-side end of the shaft in the square-marked row	STAND_ROW4_RIGHT	BEARING

#	Referring expression	Correct answer	Model said
#271	a small silver screw with a recessed Phillips head	SCREW_ROW4	SCREW_ROW1
#278	Phillips screw	SCREW_ROW4	SCREW_ROW1

2. Invalid 1 — could not be matched to any of the 26 allowed labels

#	Referring expression	Correct answer	Raw model output
#257	right bracket of the blue square row	STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT

3. Row/side over-specification 30 — answered a specific identifier (e.g. BEARING_ROW2_LEFT) instead of the grouped label (BEARING/PIN/SCREW_ROW{n})

#	Referring expression	Correct answer	Raw model output
#18	black bearing	BEARING	BEARING_ROW2_LEFT
#79	a Row 1 H5 hex-socket mounting screw	SCREW_ROW1	SCREW_ROW1_LEFT
#80	a small silver screw with a recessed H5 hex-socket head	SCREW_ROW1	SCREW_ROW1_LEFT
#81	a screw at the base of a Row 1 gear stand	SCREW_ROW1	SCREW_ROW1_LEFT
#82	a fastener used to secure a Row 1 bearing stand to the baseboard	SCREW_ROW1	SCREW_ROW1_RIGHT
#83	the silver H5 hex-socket screw for the white Row 1 stand	SCREW_ROW1	SCREW_ROW1_LEFT
#84	the Row 1 fastener tightened with a 5 mm hex key	SCREW_ROW1	SCREW_ROW1_RIGHT
#85	the small silver screw located at the foot of a white support bracket	SCREW_ROW1	SCREW_ROW1_LEFT
#86	Row 1 screw	SCREW_ROW1	SCREW_ROW1_RIGHT
#87	H5 screw	SCREW_ROW1	SCREW_ROW1_LEFT
#148	a screw at the base of a Row 2 gear stand	SCREW_ROW2	SCREW_ROW2_LEFT
#149	a fastener used to secure a Row 2 bearing stand to the baseboard	SCREW_ROW2	SCREW_ROW2_LEFT
#150	the black T25 Torx screw for the red Row 2 stand	SCREW_ROW2	SCREW_ROW2_RIGHT
#151	the Row 2 fastener tightened with a T25 driver	SCREW_ROW2	SCREW_ROW2_RIGHT
#152	the small black screw located at the foot of a red support bracket	SCREW_ROW2	SCREW_ROW2_LEFT
#213	a Row 3 H3 hex-socket mounting screw	SCREW_ROW3	SCREW_ROW3_LEFT
#214	a small black screw with a recessed H3 hex-socket head	SCREW_ROW3	SCREW_ROW3_LEFT
#215	a screw at the base of a Row 3 gear stand	SCREW_ROW3	SCREW_ROW3_LEFT
#216	a fastener used to secure a Row 3 bearing stand to the baseboard	SCREW_ROW3	SCREW_ROW3_RIGHT
#217	the black H3 hex-socket screw for the green Row 3 stand	SCREW_ROW3	SCREW_ROW3_LEFT

#	Referring expression	Correct answer	Raw model output
#218	the Row 3 fastener tightened with a 3 mm hex key	SCREW_ROW3	SCREW_ROW3_RIGHT
#219	the small black screw located at the foot of a green support bracket	SCREW_ROW3	SCREW_ROW3_LEFT
#221	H3 screw	SCREW_ROW3	SCREW_ROW3_LEFT
#270	a Row 4 Phillips mounting screw	SCREW_ROW4	SCREW_ROW4_LEFT
#272	a screw at the base of a Row 4 gear stand	SCREW_ROW4	SCREW_ROW4_LEFT
#273	a fastener used to secure a Row 4 bearing stand to the baseboard	SCREW_ROW4	SCREW_ROW4_RIGHT
#274	the silver Phillips screw for the blue Row 4 stand	SCREW_ROW4	SCREW_ROW4_RIGHT
#275	the Row 4 fastener tightened with a Phillips screwdriver	SCREW_ROW4	SCREW_ROW4_RIGHT
#276	the small silver screw located at the foot of a blue support bracket	SCREW_ROW4	SCREW_ROW4_LEFT
#277	Row 4 screw	SCREW_ROW4	SCREW_ROW4_LEFT

4. Correct, but blocked by reasoning chain 0 — the model identified the right part, but the raw output still contains a <think> block or other wrapping, so strict grading (which compares the full raw text) can never match

None.

qwen3-vl-4b (Qwen/Qwen3-VL-4B-Instruct · lenient 89.2% / strict 82.6%)

1. Valid but wrong 30 — answered a real label from the list, but not the correct one

#	Referring expression	Correct answer	Model said
#45	the left bearing support bracket in Row 1	STAND_ROW1_LEFT	BEARING
#46	the Row 1 support bracket where the left bearing fits, next to the large left gear	STAND_ROW1_LEFT	BEARING
#53	white left gear stand in the circle row	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#58	the stand immediately to the left of the crank handle	STAND_ROW1_RIGHT	STAND_ROW1_LEFT
#59	the white right bearing bracket whose bearing recess overlaps a hole through the stand	STAND_ROW1_RIGHT	BEARING
#65	white right gear stand in the circle row	STAND_ROW1_RIGHT	GEAR_ROW2_RIGHT
#102	the left bearing support bracket in the second row from Row 1	STAND_ROW2_LEFT	BEARING
#103	the Row 2 support bracket where the left bearing fits, next to the small left gear	STAND_ROW2_LEFT	BEARING
#106	the upright red bearing bracket on the left, beside the small left gear in the triangle-marked row	STAND_ROW2_LEFT	BEARING
#107	red left stand	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#110	red left gear stand in the triangle row	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#114	the right bearing support bracket in the second row from Row 1	STAND_ROW2_RIGHT	BEARING
#116	the red triangle-row right gear stand supporting the gear shaft	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#118	the upright red bearing bracket on the right, beside the medium right gear in the triangle-marked row	STAND_ROW2_RIGHT	BEARING
#122	red right gear stand in the triangle row	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#169	the left bearing support bracket in Row 3	STAND_ROW3_LEFT	BEARING
#170	the Row 3 support bracket where the left bearing fits, next to the small left gear	STAND_ROW3_LEFT	BEARING
#171	the green hexagon-row left gear stand supporting the gear shaft	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#173	the upright green bearing bracket on the left, beside the small left gear in the hexagon-marked row	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#174	green left stand	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#177	green left gear stand in the hexagon row	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#181	the right bearing support bracket in Row 3	STAND_ROW3_RIGHT	BEARING

#	Referring expression	Correct answer	Model said
#183	the green hexagon-row right gear stand supporting the gear shaft	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#186	green right stand	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#189	green right gear stand in the hexagon row	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#209	green larger gear	GEAR_ROW3_RIGHT	GEAR_ROW3_LEFT
#236	the left bearing support bracket in Row 4	STAND_ROW4_LEFT	BEARING
#237	the Row 4 support bracket where the left bearing fits, next to the large left gear	STAND_ROW4_LEFT	BEARING
#244	blue left gear stand in the square row	STAND_ROW4_LEFT	GEAR_ROW4_LEFT
#248	the right bearing support bracket in Row 4	STAND_ROW4_RIGHT	BEARING

2. Invalid 1 — could not be matched to any of the 26 allowed labels

#	Referring expression	Correct answer	Raw model output
#256	blue right gear stand in the square row	STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT

3. Row/side over-specification 19 — answered a specific identifier (e.g. BEARING_ROW2_LEFT) instead of the grouped label (BEARING/PIN/SCREW_ROW{n})

#	Referring expression	Correct answer	Raw model output
#81	a screw at the base of a Row 1 gear stand	SCREW_ROW1	SCREW_ROW1_RIGHT
#82	a fastener used to secure a Row 1 bearing stand to the baseboard	SCREW_ROW1	SCREW_ROW1_RIGHT
#83	the silver H5 hex-socket screw for the white Row 1 stand	SCREW_ROW1	SCREW_ROW1_RIGHT
#84	the Row 1 fastener tightened with a 5 mm hex key	SCREW_ROW1	SCREW_ROW1_RIGHT
#85	the small silver screw located at the foot of a white support bracket	SCREW_ROW1	SCREW_ROW1_LEFT
#150	the black T25 Torx screw for the red Row 2 stand	SCREW_ROW2	SCREW_ROW2_RIGHT
#151	the Row 2 fastener tightened with a T25 driver	SCREW_ROW2	SCREW_ROW2_RIGHT
#152	the small black screw located at the foot of a red support bracket	SCREW_ROW2	SCREW_ROW2_RIGHT
#214	a small black screw with a recessed H3 hex-socket head	SCREW_ROW3	SCREW_ROW3_LEFT
#215	a screw at the base of a Row 3 gear stand	SCREW_ROW3	SCREW_ROW3_LEFT
#217	the black H3 hex-socket screw for the green Row 3 stand	SCREW_ROW3	SCREW_ROW3_LEFT
#218	the Row 3 fastener tightened with a 3 mm hex key	SCREW_ROW3	SCREW_ROW3_LEFT
#219	the small black screw located at the foot of a green support bracket	SCREW_ROW3	SCREW_ROW3_LEFT

#	Referring expression	Correct answer	Raw model output
#271	a small silver screw with a recessed Phillips head	SCREW_ROW4	SCREW_ROW4_RIGHT
#272	a screw at the base of a Row 4 gear stand	SCREW_ROW4	SCREW_ROW4_LEFT
#274	the silver Phillips screw for the blue Row 4 stand	SCREW_ROW4	SCREW_ROW4_RIGHT
#275	the Row 4 fastener tightened with a Phillips screwdriver	SCREW_ROW4	SCREW_ROW4_RIGHT
#276	the small silver screw located at the foot of a blue support bracket	SCREW_ROW4	SCREW_ROW4_LEFT
#278	Phillips screw	SCREW_ROW4	SCREW_ROW4_RIGHT

4. Correct, but blocked by reasoning chain 0 — the model identified the right part, but the raw output still contains a <think> block or other wrapping, so strict grading (which compares the full raw text) can never match
None.

qwen3-8b (Qwen/Qwen3-8B · lenient 86.8% / strict 0.0%)

1. Valid but wrong 0 — answered a real label from the list, but not the correct one

None.

2. Invalid 38 — could not be matched to any of the 26 allowed labels

#	Referring expression	Correct answer	Raw model output
#16	the black bearing that supports a rotating gear shaft inside a stand	BEARING	<think> Okay, let's tackle this problem. The user is asking for the single physical part referred to by the expression "the black bearing that supports a rotating gear shaft inside a stand." First, I need to recall the cut off mid-reasoning, no answer reached
#18	black bearing	BEARING	<think> Okay, let's tackle this problem. The user's referring expression is "black bearing." I need to figure out which single physical part this refers to based on the provided task description and allowed labels. Firs cut off mid-reasoning, no answer reached
#23	the handle immediately to the right of the right white gear stand	CRANK_HANDLE_ROW1	<think> Okay, let's tackle this problem. The user is asking for the single physical part referred to by the expression "the handle immediately to the right of the right white gear stand." First, I need to parse the exp cut off mid-reasoning, no answer reached
#30	right-side crank	CRANK_HANDLE_ROW1	<think> Okay, let's tackle this problem. The user's referring expression is "right-side crank." I need to figure out which single physical part this refers to based on the task description and allowed labels. First, I'l cut off mid-reasoning, no answer reached
#57	the right-hand support in circle-marked Row 1	STAND_ROW1_RIGHT	<think> Okay, let's tackle this problem. The user's expression is "the right-hand support in circle-marked Row 1". I need to figure out which single physical part this refers to based on the task description and allowed cut off mid-reasoning, no answer reached
#58	the stand immediately to the left of the crank handle	STAND_ROW1_RIGHT	<think> Okay, let's tackle this problem. The user is asking for the physical part referred to by the expression "the stand immediately to the left of the crank handle." First, I need to recall the task description and cut off mid-reasoning, no answer reached

#	Referring expression	Correct answer	Raw model output
#66	right bracket of the white circle row	STAND_ROW1_RIGHT	<think> Okay, let's tackle this problem. The user is asking about the referring expression "right bracket of the white circle row." I need to figure out which single physical part this refers to based on the task descrip cut off mid-reasoning, no answer reached
#76	Row 1 gear	GEAR_ROW1_LEFT	<think> Okay, let's tackle this problem. The user is asking about the referring expression "Row 1 gear." I need to figure out which single physical part this refers to based on the task description and allowed labels. F cut off mid-reasoning, no answer reached
#78	circle-row white gear	GEAR_ROW1_LEFT	<think> Okay, let's tackle this problem. The user's referring expression is "circle-row white gear". I need to figure out which single physical part this refers to based on the task description and allowed labels. First cut off mid-reasoning, no answer reached
#85	the small silver screw located at the foot of a white support bracket	SCREW_ROW1	<think> Okay, let's tackle this problem. The user is asking for the single physical part referred to by the expression: "the small silver screw located at the foot of a white support bracket." First, I need to parse th cut off mid-reasoning, no answer reached
#102	the left bearing support bracket in the second row from Row 1	STAND_ROW2_LEFT	<think> Okay, let's tackle this problem. The user is asking for the single physical part referred to by the expression "the left bearing support bracket in the second row from Row 1". First, I need to parse the express cut off mid-reasoning, no answer reached
#106	the upright red bearing bracket on the left, beside the small left gear in the triangle-marked row	STAND_ROW2_LEFT	<think> Okay, let's break down the user's expression: "the upright red bearing bracket on the left, beside the small left gear in the triangle-marked row." First, the color and shape mention. The user refers to the "tr cut off mid-reasoning, no answer reached
#113	the upright red right stand with a circular bearing opening	STAND_ROW2_RIGHT	<think> Okay, let's tackle this problem. The user is asking for the single physical part referred to by the expression "the upright red right stand with a circular bearing opening." First, I need to parse the descripti cut off mid-reasoning, no answer reached

#	Referring expression	Correct answer	Raw model output
#123	right bracket of the red triangle row	STAND_ROW2_RIGHT	<think> Okay, let's tackle this problem. The user is asking about the referring expression "right bracket of the red triangle row." I need to figure out which single physical part this refers to based on the task descrip cut off mid-reasoning, no answer reached
#131	red smaller gear	GEAR_ROW2_LEFT	<think> Okay, let's tackle this problem. The user's referring expression is "red smaller gear." I need to figure out which single physical part this refers to based on the task description and allowed labels. First, let cut off mid-reasoning, no answer reached
#140	the larger red gear to the right of the small red gear	GEAR_ROW2_RIGHT	<think> Okay, let's tackle this problem. The user is asking about the referring expression: "the larger red gear to the right of the small red gear." I need to figure out which single physical part this refers to based o cut off mid-reasoning, no answer reached
#142	red larger gear	GEAR_ROW2_RIGHT	<think> Okay, let's tackle this problem. The user's referring expression is "red larger gear." I need to figure out which single physical part this refers to based on the task description and allowed labels. First, let' cut off mid-reasoning, no answer reached
#148	a screw at the base of a Row 2 gear stand	SCREW_ROW2	<think> Okay, let's tackle this problem. The user is asking for the single physical part referred to by the expression "a screw at the base of a Row 2 gear stand." First, I need to recall the allowed dataset labels and cut off mid-reasoning, no answer reached
#185	the upright green bearing bracket on the right, beside the medium right gear in the hexagon-marked row	STAND_ROW3_RIGHT	<think> Okay, let's break down the user's expression: "the upright green bearing bracket on the right, beside the medium right gear in the hexagon-marked row." First, the color and shape mention. The hexagon-marked row cut off mid-reasoning, no answer reached
#207	the larger green gear to the right of the small green gear	GEAR_ROW3_RIGHT	<think> Okay, let's tackle this problem. The user is asking about the referring expression: "the larger green gear to the right of the small green gear." I need to figure out which single physical part this refers to bas cut off mid-reasoning, no answer reached

#	Referring expression	Correct answer	Raw model output
#209	green larger gear	GEAR_ROW3_RIGHT	<think> Okay, let's tackle this problem. The user's referring expression is "green larger gear." I need to figure out which single physical part this refers to based on the task description and allowed labels. First, le cut off mid-reasoning, no answer reached
#211	larger green gear in the hexagon row	GEAR_ROW3_RIGHT	<think> Okay, let's tackle this problem. The user is asking about the referring expression "larger green gear in the hexagon row." I need to figure out which specific part this refers to based on the task description and cut off mid-reasoning, no answer reached
#240	the upright blue bearing bracket on the left, beside the large left gear in the square-marked row	STAND_ROW4_LEFT	<think> Okay, let's break down the user's expression: "the upright blue bearing bracket on the left, beside the large left gear in the square-marked row." First, the color and shape mention. The user says "blue and squ cut off mid-reasoning, no answer reached
#252	the upright blue bearing bracket on the right, beside the crank-side end of the shaft in the square-marked row	STAND_ROW4_RIGHT	<think> Okay, let's break down the user's expression: "the upright blue bearing bracket on the right, beside the crank-side end of the shaft in the square-marked row." First, the color mentioned is blue. According to th cut off mid-reasoning, no answer reached
#257	right bracket of the blue square row	STAND_ROW4_RIGHT	<think> Okay, let's tackle this problem. The user is asking about the referring expression "right bracket of the blue square row." I need to figure out which single physical part this refers to based on the task descript cut off mid-reasoning, no answer reached
#259	the large blue spur wheel	GEAR_ROW4_LEFT	<think> Okay, let's tackle this problem. The user is asking about the referring expression "the large blue spur wheel." I need to figure out which single physical part this refers to based on the provided task descriptio cut off mid-reasoning, no answer reached
#266	blue gear	GEAR_ROW4_LEFT	<think> Okay, let's tackle this problem. The user's referring expression is "blue gear." I need to figure out which single physical part this refers to based on the task description and allowed labels. First, I'll recal cut off mid-reasoning, no answer reached

#	Referring expression	Correct answer	Raw model output
#267	Row 4 gear	GEAR_ROW4_LEFT	<think> Okay, let's tackle this problem. The user is asking about the referring expression "Row 4 gear." I need to figure out which single physical part this refers to based on the task description and allowed labels. F cut off mid-reasoning, no answer reached
#268	blue gear in the square row	GEAR_ROW4_LEFT	<think> Okay, let's tackle this problem. The user's referring expression is "blue gear in the square row." I need to figure out which single physical part this refers to based on the task description and allowed labels. cut off mid-reasoning, no answer reached
#269	square-row blue gear	GEAR_ROW4_LEFT	<think> Okay, let's tackle this problem. The user's referring expression is "square-row blue gear." I need to figure out which single physical part this refers to based on the task description and allowed labels. First, cut off mid-reasoning, no answer reached
#271	a small silver screw with a recessed Phillips head	SCREW_ROW4	<think> Okay, let's tackle this problem. The user is asking to identify the single physical part referred to by the expression "a small silver screw with a recessed Phillips head." First, I need to look at the allowed cut off mid-reasoning, no answer reached
#272	a screw at the base of a Row 4 gear stand	SCREW_ROW4	<think> Okay, let's tackle this problem. The user is asking for the single physical part referred to by the expression "a screw at the base of a Row 4 gear stand." First, I need to recall the allowed dataset labels and cut off mid-reasoning, no answer reached
#273	a fastener used to secure a Row 4 bearing stand to the baseboard	SCREW_ROW4	<think> Okay, let's tackle this problem. The user is asking for the single physical part referred to by the expression "a fastener used to secure a Row 4 bearing stand to the baseboard." First, I need to recall the all cut off mid-reasoning, no answer reached
#276	the small silver screw located at the foot of a blue support bracket	SCREW_ROW4	<think> Okay, let's tackle this problem. The user is asking for the single physical part referred to by the expression: "the small silver screw located at the foot of a blue support bracket." First, I need to parse the cut off mid-reasoning, no answer reached

#	Referring expression	Correct answer	Raw model output
#280	the short thin brown cylindrical peg	PIN	<think> Okay, let's tackle this problem. The user is asking to identify the single physical part referred to by the expression "the short thin brown cylindrical peg." First, I need to recall the allowed dataset labels cut off mid-reasoning, no answer reached
#283	the small brown cylindrical retaining pin fitted beside a gear	PIN	<think> Okay, let's tackle this problem. The user is asking to identify the single physical part referred to by the expression "the small brown cylindrical retaining pin fitted beside a gear." First, I need to recall t cut off mid-reasoning, no answer reached
#285	the short brown dowel crossing the gear rod beside a gear or the crank	PIN	<think> Okay, let's tackle this problem. The user is asking about the referring expression: "the short brown dowel crossing the gear rod beside a gear or the crank." I need to figure out which single physical part this r cut off mid-reasoning, no answer reached
#287	brown pin	PIN	<think> Okay, let's tackle this problem. The user's referring expression is "brown pin." I need to figure out which single physical part this refers to based on the provided task description and allowed labels. First, I cut off mid-reasoning, no answer reached

3. Row/side over-specification 0 — answered a specific identifier (e.g. BEARING_ROW2_LEFT) instead of the grouped label (BEARING/PIN/SCREW_ROW{n})

None.

4. Correct, but blocked by reasoning chain 250 — the model identified the right part, but the raw output still contains a <think> block or other wrapping, so strict grading (which compares the full raw text) can never match

#	Referring expression	Correct answer	Raw model output
#1	the gearbox mounting plate	BASE_BOARD	BASE_BOARD had <think>
#2	the large flat black rectangular plate	BASE_BOARD	BASE_BOARD had <think>
#3	the base plate underneath all four colored rows	BASE_BOARD	BASE_BOARD had <think>
#4	the plate to which all eight bearing support brackets are screwed	BASE_BOARD	BASE_BOARD had <think>
#5	the large black rectangular mounting plate below every gear shaft and stand	BASE_BOARD	BASE_BOARD had <think>
#6	the rigid plate that supports and locates every gearbox row	BASE_BOARD	BASE_BOARD had <think>
#7	the broad black rectangular base beneath the white, red, green, and blue mechanisms	BASE_BOARD	BASE_BOARD had <think>
#8	black baseboard	BASE_BOARD	BASE_BOARD had <think>
#9	gearbox base	BASE_BOARD	BASE_BOARD had <think>

#	Referring expression	Correct answer	Raw model output
#10	mounting plate	BASE_BOARD	BASE_BOARD had <think>
#11	the radial shaft-support bearing	BEARING	BEARING had <think>
#12	the small black circular ring with a center hole	BEARING	BEARING had <think>
#13	the round black part centered inside a bearing support bracket	BEARING	BEARING had <think>
#14	the bearing through which a gear shaft passes	BEARING	BEARING had <think>
#15	the small black ring-shaped bearing that fits into each colored stand	BEARING	BEARING had <think>
#17	the small black ring seated in the round opening of a colored support bracket	BEARING	BEARING had <think>
#19	circular bearing	BEARING	BEARING had <think>
#20	gear-stand bearing	BEARING	BEARING had <think>
#21	the Row 1 hand crank	CRANK_HANDLE_ROW1	CRANK_HANDLE_ROW1 had <think>
#22	the white bent lever with a rounded grip	CRANK_HANDLE_ROW1	CRANK_HANDLE_ROW1 had <think>
#24	the crank attached at the right end of the white gear rod	CRANK_HANDLE_ROW1	CRANK_HANDLE_ROW1 had <think>
#25	the white handle beside the right stand in circle-marked Row 1	CRANK_HANDLE_ROW1	CRANK_HANDLE_ROW1 had <think>
#26	the hand-operated lever used to rotate the assembled gear train	CRANK_HANDLE_ROW1	CRANK_HANDLE_ROW1 had <think>
#27	the bent white crank connected to the end of the Row 1 shaft	CRANK_HANDLE_ROW1	CRANK_HANDLE_ROW1 had <think>
#28	white handle	CRANK_HANDLE_ROW1	CRANK_HANDLE_ROW1 had <think>
#29	Row 1 crank	CRANK_HANDLE_ROW1	CRANK_HANDLE_ROW1 had <think>
#31	the Row 1 gear shaft	GEAR_ROD_ROW1	GEAR_ROD_ROW1 had <think>
#32	the long white cylindrical shaft with two pin holes	GEAR_ROD_ROW1	GEAR_ROD_ROW1 had <think>
#33	the shaft crossing Row 1 between the two white brackets	GEAR_ROD_ROW1	GEAR_ROD_ROW1 had <think>
#34	the shaft that passes through both Row 1 bearings and carries one large gear	GEAR_ROD_ROW1	GEAR_ROD_ROW1 had <think>
#35	the long white circle-row gear shaft seated in both bearing brackets	GEAR_ROD_ROW1	GEAR_ROD_ROW1 had <think>
#36	the Row 1 shaft that carries one large gear and transfers rotation between the two stands	GEAR_ROD_ROW1	GEAR_ROD_ROW1 had <think>
#37	the long white cylindrical rod with two retaining-pin holes spanning Row 1	GEAR_ROD_ROW1	GEAR_ROD_ROW1 had <think>
#38	white gear rod	GEAR_ROD_ROW1	GEAR_ROD_ROW1 had <think>
#39	Row 1 shaft	GEAR_ROD_ROW1	GEAR_ROD_ROW1 had <think>
#40	gear rod of Row 1	GEAR_ROD_ROW1	GEAR_ROD_ROW1 had <think>
#41	white gear rod in the circle row	GEAR_ROD_ROW1	GEAR_ROD_ROW1 had <think>
#42	circle-row white shaft	GEAR_ROD_ROW1	GEAR_ROD_ROW1 had <think>
#43	the Row 1 left gear stand	STAND_ROW1_LEFT	STAND_ROW1_LEFT had <think>

#	Referring expression	Correct answer	Raw model output
#44	the upright white left stand with a circular bearing opening	STAND_ROW1_LEFT	STAND_ROW1_LEFT had <think>
#45	the left bearing support bracket in Row 1	STAND_ROW1_LEFT	STAND_ROW1_LEFT had <think>
#46	the Row 1 support bracket where the left bearing fits, next to the large left gear	STAND_ROW1_LEFT	STAND_ROW1_LEFT had <think>
#47	the white circle-row left gear stand supporting the gear shaft	STAND_ROW1_LEFT	STAND_ROW1_LEFT had <think>
#48	the left Row 1 support that holds a bearing and supports the gear shaft	STAND_ROW1_LEFT	STAND_ROW1_LEFT had <think>
#49	the upright white bearing bracket on the left, beside the large left gear in the circle-marked row	STAND_ROW1_LEFT	STAND_ROW1_LEFT had <think>
#50	white left stand	STAND_ROW1_LEFT	STAND_ROW1_LEFT had <think>
#51	Row 1 left bracket	STAND_ROW1_LEFT	STAND_ROW1_LEFT had <think>
#52	left gear stand of Row 1	STAND_ROW1_LEFT	STAND_ROW1_LEFT had <think>
#53	white left gear stand in the circle row	STAND_ROW1_LEFT	STAND_ROW1_LEFT had <think>
#54	left bracket of the white circle row	STAND_ROW1_LEFT	STAND_ROW1_LEFT had <think>
#55	the Row 1 right gear stand	STAND_ROW1_RIGHT	STAND_ROW1_RIGHT had <think>
#56	the upright white stand with an overlapping bearing slot and through-hole	STAND_ROW1_RIGHT	STAND_ROW1_RIGHT had <think>
#59	the white right bearing bracket whose bearing recess overlaps a hole through the stand	STAND_ROW1_RIGHT	STAND_ROW1_RIGHT had <think>
#60	the right Row 1 support that holds the bearing and guides the gear shaft	STAND_ROW1_RIGHT	STAND_ROW1_RIGHT had <think>
#61	the white crank-side gear stand with both a bearing slot and a penetrating hole	STAND_ROW1_RIGHT	STAND_ROW1_RIGHT had <think>
#62	white right stand	STAND_ROW1_RIGHT	STAND_ROW1_RIGHT had <think>
#63	Row 1 right bracket	STAND_ROW1_RIGHT	STAND_ROW1_RIGHT had <think>
#64	right gear stand of Row 1	STAND_ROW1_RIGHT	STAND_ROW1_RIGHT had <think>
#65	white right gear stand in the circle row	STAND_ROW1_RIGHT	STAND_ROW1_RIGHT had <think>
#67	the Row 1 left spur gear	GEAR_ROW1_LEFT	GEAR_ROW1_LEFT had <think>
#68	the large white spur wheel	GEAR_ROW1_LEFT	GEAR_ROW1_LEFT had <think>
#69	the left-side gear on the shaft in Row 1	GEAR_ROW1_LEFT	GEAR_ROW1_LEFT had <think>
#70	the Row 1 gear next to the left white bracket, the only gear in Row 1	GEAR_ROW1_LEFT	GEAR_ROW1_LEFT had <think>
#71	the large white spur gear on the left side of the circle-marked shaft	GEAR_ROW1_LEFT	GEAR_ROW1_LEFT had <think>
#72	the Row 1 spur gear that transmits rotation along the gear train	GEAR_ROW1_LEFT	GEAR_ROW1_LEFT had <think>
#73	the large white toothed wheel on the left side, beside the gear shaft	GEAR_ROW1_LEFT	GEAR_ROW1_LEFT had <think>
#74	gear of Row 1	GEAR_ROW1_LEFT	GEAR_ROW1_LEFT had <think>

#	Referring expression	Correct answer	Raw model output
#75	white gear	GEAR_ROW1_LEFT	GEAR_ROW1_LEFT had <think>
#77	white gear in the circle row	GEAR_ROW1_LEFT	GEAR_ROW1_LEFT had <think>
#79	a Row 1 H5 hex-socket mounting screw	SCREW_ROW1	SCREW_ROW1 had <think>
#80	a small silver screw with a recessed H5 hex-socket head	SCREW_ROW1	SCREW_ROW1 had <think>
#81	a screw at the base of a Row 1 gear stand	SCREW_ROW1	SCREW_ROW1 had <think>
#82	a fastener used to secure a Row 1 bearing stand to the baseboard	SCREW_ROW1	SCREW_ROW1 had <think>
#83	the silver H5 hex-socket screw for the white Row 1 stand	SCREW_ROW1	SCREW_ROW1 had <think>
#84	the Row 1 fastener tightened with a 5 mm hex key	SCREW_ROW1	SCREW_ROW1 had <think>
#86	Row 1 screw	SCREW_ROW1	SCREW_ROW1 had <think>
#87	H5 screw	SCREW_ROW1	SCREW_ROW1 had <think>
#88	the Row 2 gear shaft	GEAR_ROD_ROW2	GEAR_ROD_ROW2 had <think>
#89	the long red cylindrical shaft with two pin holes	GEAR_ROD_ROW2	GEAR_ROD_ROW2 had <think>
#90	the shaft crossing Row 2 between the two red brackets	GEAR_ROD_ROW2	GEAR_ROD_ROW2 had <think>
#91	the shaft that passes through both Row 2 bearings and carries the small and medium gears	GEAR_ROD_ROW2	GEAR_ROD_ROW2 had <think>
#92	the long red triangle-row gear shaft seated in both bearing brackets	GEAR_ROD_ROW2	GEAR_ROD_ROW2 had <think>
#93	the Row 2 shaft that carries the small and medium gears and transfers rotation between the two stands	GEAR_ROD_ROW2	GEAR_ROD_ROW2 had <think>
#94	the long red cylindrical rod with two retaining-pin holes spanning the second row from Row 1	GEAR_ROD_ROW2	GEAR_ROD_ROW2 had <think>
#95	red gear rod	GEAR_ROD_ROW2	GEAR_ROD_ROW2 had <think>
#96	Row 2 shaft	GEAR_ROD_ROW2	GEAR_ROD_ROW2 had <think>
#97	gear rod of Row 2	GEAR_ROD_ROW2	GEAR_ROD_ROW2 had <think>
#98	red gear rod in the triangle row	GEAR_ROD_ROW2	GEAR_ROD_ROW2 had <think>
#99	triangle-row red shaft	GEAR_ROD_ROW2	GEAR_ROD_ROW2 had <think>
#100	the Row 2 left gear stand	STAND_ROW2_LEFT	STAND_ROW2_LEFT had <think>
#101	the upright red left stand with a circular bearing opening	STAND_ROW2_LEFT	STAND_ROW2_LEFT had <think>
#103	the Row 2 support bracket where the left bearing fits, next to the small left gear	STAND_ROW2_LEFT	STAND_ROW2_LEFT had <think>
#104	the red triangle-row left gear stand supporting the gear shaft	STAND_ROW2_LEFT	STAND_ROW2_LEFT had <think>
#105	the left Row 2 support that holds a bearing and supports the gear shaft	STAND_ROW2_LEFT	STAND_ROW2_LEFT had <think>
#107	red left stand	STAND_ROW2_LEFT	STAND_ROW2_LEFT had <think>

#	Referring expression	Correct answer	Raw model output
#108	Row 2 left bracket	STAND_ROW2_LEFT	STAND_ROW2_LEFT had <think>
#109	left gear stand of Row 2	STAND_ROW2_LEFT	STAND_ROW2_LEFT had <think>
#110	red left gear stand in the triangle row	STAND_ROW2_LEFT	STAND_ROW2_LEFT had <think>
#111	left bracket of the red triangle row	STAND_ROW2_LEFT	STAND_ROW2_LEFT had <think>
#112	the Row 2 right gear stand	STAND_ROW2_RIGHT	STAND_ROW2_RIGHT had <think>
#114	the right bearing support bracket in the second row from Row 1	STAND_ROW2_RIGHT	STAND_ROW2_RIGHT had <think>
#115	the Row 2 support bracket where the right bearing fits, next to the medium right gear	STAND_ROW2_RIGHT	STAND_ROW2_RIGHT had <think>
#116	the red triangle-row right gear stand supporting the gear shaft	STAND_ROW2_RIGHT	STAND_ROW2_RIGHT had <think>
#117	the right Row 2 support that holds a bearing and supports the gear shaft	STAND_ROW2_RIGHT	STAND_ROW2_RIGHT had <think>
#118	the upright red bearing bracket on the right, beside the medium right gear in the triangle-marked row	STAND_ROW2_RIGHT	STAND_ROW2_RIGHT had <think>
#119	red right stand	STAND_ROW2_RIGHT	STAND_ROW2_RIGHT had <think>
#120	Row 2 right bracket	STAND_ROW2_RIGHT	STAND_ROW2_RIGHT had <think>
#121	right gear stand of Row 2	STAND_ROW2_RIGHT	STAND_ROW2_RIGHT had <think>
#122	red right gear stand in the triangle row	STAND_ROW2_RIGHT	STAND_ROW2_RIGHT had <think>
#124	the Row 2 left spur gear	GEAR_ROW2_LEFT	GEAR_ROW2_LEFT had <think>
#125	the small red spur wheel	GEAR_ROW2_LEFT	GEAR_ROW2_LEFT had <think>
#126	the left-side gear on the shaft in the second row from Row 1	GEAR_ROW2_LEFT	GEAR_ROW2_LEFT had <think>
#127	the Row 2 gear next to the left red bracket, smaller than the right gear	GEAR_ROW2_LEFT	GEAR_ROW2_LEFT had <think>
#128	the small red spur gear on the left side of the triangle-marked shaft	GEAR_ROW2_LEFT	GEAR_ROW2_LEFT had <think>
#129	the small red gear to the left of the larger red gear	GEAR_ROW2_LEFT	GEAR_ROW2_LEFT had <think>
#130	smaller gear of Row 2	GEAR_ROW2_LEFT	GEAR_ROW2_LEFT had <think>
#132	left gear of Row 2	GEAR_ROW2_LEFT	GEAR_ROW2_LEFT had <think>
#133	smaller red gear in the triangle row	GEAR_ROW2_LEFT	GEAR_ROW2_LEFT had <think>
#134	triangle-row red left gear	GEAR_ROW2_LEFT	GEAR_ROW2_LEFT had <think>
#135	the Row 2 right spur gear	GEAR_ROW2_RIGHT	GEAR_ROW2_RIGHT had <think>
#136	the medium red spur wheel	GEAR_ROW2_RIGHT	GEAR_ROW2_RIGHT had <think>
#137	the right-side gear on the shaft in the second row from Row 1	GEAR_ROW2_RIGHT	GEAR_ROW2_RIGHT had <think>
#138	the Row 2 gear next to the right red bracket, larger than the left gear	GEAR_ROW2_RIGHT	GEAR_ROW2_RIGHT had <think>
#139	the medium red spur gear on the right side of the triangle-marked shaft	GEAR_ROW2_RIGHT	GEAR_ROW2_RIGHT had <think>
#141	larger gear of Row 2	GEAR_ROW2_RIGHT	GEAR_ROW2_RIGHT had <think>
#143	right gear of Row 2	GEAR_ROW2_RIGHT	GEAR_ROW2_RIGHT had <think>
#144	larger red gear in the triangle row	GEAR_ROW2_RIGHT	GEAR_ROW2_RIGHT had <think>

#	Referring expression	Correct answer	Raw model output
#145	triangle-row red right gear	GEAR_ROW2_RIGHT	GEAR_ROW2_RIGHT had <think>
#146	a Row 2 T25 Torx mounting screw	SCREW_ROW2	SCREW_ROW2 had <think>
#147	a small black screw with a recessed T25 Torx head	SCREW_ROW2	SCREW_ROW2 had <think>
#149	a fastener used to secure a Row 2 bearing stand to the baseboard	SCREW_ROW2	SCREW_ROW2 had <think>
#150	the black T25 Torx screw for the red Row 2 stand	SCREW_ROW2	SCREW_ROW2 had <think>
#151	the Row 2 fastener tightened with a T25 driver	SCREW_ROW2	SCREW_ROW2 had <think>
#152	the small black screw located at the foot of a red support bracket	SCREW_ROW2	SCREW_ROW2 had <think>
#153	Row 2 screw	SCREW_ROW2	SCREW_ROW2 had <think>
#154	T25 screw	SCREW_ROW2	SCREW_ROW2 had <think>
#155	the Row 3 gear shaft	GEAR_ROD_ROW3	GEAR_ROD_ROW3 had <think>
#156	the long green cylindrical shaft with two pin holes	GEAR_ROD_ROW3	GEAR_ROD_ROW3 had <think>
#157	the shaft crossing Row 3 between the two green brackets	GEAR_ROD_ROW3	GEAR_ROD_ROW3 had <think>
#158	the shaft that passes through both Row 3 bearings and carries the small and medium gears	GEAR_ROD_ROW3	GEAR_ROD_ROW3 had <think>
#159	the long green hexagon-row gear shaft seated in both bearing brackets	GEAR_ROD_ROW3	GEAR_ROD_ROW3 had <think>
#160	the Row 3 shaft that carries the small and medium gears and transfers rotation between the two stands	GEAR_ROD_ROW3	GEAR_ROD_ROW3 had <think>
#161	the long green cylindrical rod with two retaining-pin holes spanning Row 3	GEAR_ROD_ROW3	GEAR_ROD_ROW3 had <think>
#162	green gear rod	GEAR_ROD_ROW3	GEAR_ROD_ROW3 had <think>
#163	Row 3 shaft	GEAR_ROD_ROW3	GEAR_ROD_ROW3 had <think>
#164	gear rod of Row 3	GEAR_ROD_ROW3	GEAR_ROD_ROW3 had <think>
#165	green gear rod in the hexagon row	GEAR_ROD_ROW3	GEAR_ROD_ROW3 had <think>
#166	hexagon-row green shaft	GEAR_ROD_ROW3	GEAR_ROD_ROW3 had <think>
#167	the Row 3 left gear stand	STAND_ROW3_LEFT	STAND_ROW3_LEFT had <think>
#168	the upright green left stand with a circular bearing opening	STAND_ROW3_LEFT	STAND_ROW3_LEFT had <think>
#169	the left bearing support bracket in Row 3	STAND_ROW3_LEFT	STAND_ROW3_LEFT had <think>
#170	the Row 3 support bracket where the left bearing fits, next to the small left gear	STAND_ROW3_LEFT	STAND_ROW3_LEFT had <think>
#171	the green hexagon-row left gear stand supporting the gear shaft	STAND_ROW3_LEFT	STAND_ROW3_LEFT had <think>
#172	the left Row 3 support that holds a bearing and supports the gear shaft	STAND_ROW3_LEFT	STAND_ROW3_LEFT had <think>

#	Referring expression	Correct answer	Raw model output
#173	the upright green bearing bracket on the left, beside the small left gear in the hexagon-marked row	STAND_ROW3_LEFT	STAND_ROW3_LEFT had <think>
#174	green left stand	STAND_ROW3_LEFT	STAND_ROW3_LEFT had <think>
#175	Row 3 left bracket	STAND_ROW3_LEFT	STAND_ROW3_LEFT had <think>
#176	left gear stand of Row 3	STAND_ROW3_LEFT	STAND_ROW3_LEFT had <think>
#177	green left gear stand in the hexagon row	STAND_ROW3_LEFT	STAND_ROW3_LEFT had <think>
#178	left bracket of the green hexagon row	STAND_ROW3_LEFT	STAND_ROW3_LEFT had <think>
#179	the Row 3 right gear stand	STAND_ROW3_RIGHT	STAND_ROW3_RIGHT had <think>
#180	the upright green right stand with a circular bearing opening	STAND_ROW3_RIGHT	STAND_ROW3_RIGHT had <think>
#181	the right bearing support bracket in Row 3	STAND_ROW3_RIGHT	STAND_ROW3_RIGHT had <think>
#182	the Row 3 support bracket where the right bearing fits, next to the medium right gear	STAND_ROW3_RIGHT	STAND_ROW3_RIGHT had <think>
#183	the green hexagon-row right gear stand supporting the gear shaft	STAND_ROW3_RIGHT	STAND_ROW3_RIGHT had <think>
#184	the right Row 3 support that holds a bearing and supports the gear shaft	STAND_ROW3_RIGHT	STAND_ROW3_RIGHT had <think>
#186	green right stand	STAND_ROW3_RIGHT	STAND_ROW3_RIGHT had <think>
#187	Row 3 right bracket	STAND_ROW3_RIGHT	STAND_ROW3_RIGHT had <think>
#188	right gear stand of Row 3	STAND_ROW3_RIGHT	STAND_ROW3_RIGHT had <think>
#189	green right gear stand in the hexagon row	STAND_ROW3_RIGHT	STAND_ROW3_RIGHT had <think>
#190	right bracket of the green hexagon row	STAND_ROW3_RIGHT	STAND_ROW3_RIGHT had <think>
#191	the Row 3 left spur gear	GEAR_ROW3_LEFT	GEAR_ROW3_LEFT had <think>
#192	the small green spur wheel	GEAR_ROW3_LEFT	GEAR_ROW3_LEFT had <think>
#193	the left-side gear on the shaft in Row 3	GEAR_ROW3_LEFT	GEAR_ROW3_LEFT had <think>
#194	the Row 3 gear next to the left green bracket, smaller than the right gear	GEAR_ROW3_LEFT	GEAR_ROW3_LEFT had <think>
#195	the small green spur gear on the left side of the hexagon-marked shaft	GEAR_ROW3_LEFT	GEAR_ROW3_LEFT had <think>
#196	the small green gear to the left of the larger green gear	GEAR_ROW3_LEFT	GEAR_ROW3_LEFT had <think>
#197	smaller gear of Row 3	GEAR_ROW3_LEFT	GEAR_ROW3_LEFT had <think>
#198	green smaller gear	GEAR_ROW3_LEFT	GEAR_ROW3_LEFT had <think>
#199	left gear of Row 3	GEAR_ROW3_LEFT	GEAR_ROW3_LEFT had <think>
#200	smaller green gear in the hexagon row	GEAR_ROW3_LEFT	GEAR_ROW3_LEFT had <think>
#201	hexagon-row green left gear	GEAR_ROW3_LEFT	GEAR_ROW3_LEFT had <think>
#202	the Row 3 right spur gear	GEAR_ROW3_RIGHT	GEAR_ROW3_RIGHT had <think>
#203	the medium green spur wheel	GEAR_ROW3_RIGHT	GEAR_ROW3_RIGHT had <think>
#204	the right-side gear on the shaft in Row 3	GEAR_ROW3_RIGHT	GEAR_ROW3_RIGHT had <think>

#	Referring expression	Correct answer	Raw model output
#205	the Row 3 gear next to the right green bracket, larger than the left gear	GEAR_ROW3_RIGHT	GEAR_ROW3_RIGHT had <think>
#206	the medium green spur gear on the right side of the hexagon-marked shaft	GEAR_ROW3_RIGHT	GEAR_ROW3_RIGHT had <think>
#208	larger gear of Row 3	GEAR_ROW3_RIGHT	GEAR_ROW3_RIGHT had <think>
#210	right gear of Row 3	GEAR_ROW3_RIGHT	GEAR_ROW3_RIGHT had <think>
#212	hexagon-row green right gear	GEAR_ROW3_RIGHT	GEAR_ROW3_RIGHT had <think>
#213	a Row 3 H3 hex-socket mounting screw	SCREW_ROW3	SCREW_ROW3 had <think>
#214	a small black screw with a recessed H3 hex-socket head	SCREW_ROW3	SCREW_ROW3 had <think>
#215	a screw at the base of a Row 3 gear stand	SCREW_ROW3	SCREW_ROW3 had <think>
#216	a fastener used to secure a Row 3 bearing stand to the baseboard	SCREW_ROW3	SCREW_ROW3 had <think>
#217	the black H3 hex-socket screw for the green Row 3 stand	SCREW_ROW3	SCREW_ROW3 had <think>
#218	the Row 3 fastener tightened with a 3 mm hex key	SCREW_ROW3	SCREW_ROW3 had <think>
#219	the small black screw located at the foot of a green support bracket	SCREW_ROW3	SCREW_ROW3 had <think>
#220	Row 3 screw	SCREW_ROW3	SCREW_ROW3 had <think>
#221	H3 screw	SCREW_ROW3	SCREW_ROW3 had <think>
#222	the Row 4 gear shaft	GEAR_ROD_ROW4	GEAR_ROD_ROW4 had <think>
#223	the long blue cylindrical shaft with one pin hole	GEAR_ROD_ROW4	GEAR_ROD_ROW4 had <think>
#224	the shaft crossing Row 4 between the two blue brackets	GEAR_ROD_ROW4	GEAR_ROD_ROW4 had <think>
#225	the shaft that passes through both Row 4 bearings and carries one large gear	GEAR_ROD_ROW4	GEAR_ROD_ROW4 had <think>
#226	the long blue square-row gear shaft seated in both bearing brackets	GEAR_ROD_ROW4	GEAR_ROD_ROW4 had <think>
#227	the Row 4 shaft that carries one large gear and transfers rotation between the two stands	GEAR_ROD_ROW4	GEAR_ROD_ROW4 had <think>
#228	the long blue cylindrical rod with one retaining-pin hole spanning Row 4	GEAR_ROD_ROW4	GEAR_ROD_ROW4 had <think>
#229	blue gear rod	GEAR_ROD_ROW4	GEAR_ROD_ROW4 had <think>
#230	Row 4 shaft	GEAR_ROD_ROW4	GEAR_ROD_ROW4 had <think>
#231	gear rod of Row 4	GEAR_ROD_ROW4	GEAR_ROD_ROW4 had <think>
#232	blue gear rod in the square row	GEAR_ROD_ROW4	GEAR_ROD_ROW4 had <think>
#233	square-row blue shaft	GEAR_ROD_ROW4	GEAR_ROD_ROW4 had <think>
#234	the Row 4 left gear stand	STAND_ROW4_LEFT	STAND_ROW4_LEFT had <think>
#235	the upright blue left stand with a circular bearing opening	STAND_ROW4_LEFT	STAND_ROW4_LEFT had <think>
#236	the left bearing support bracket in Row 4	STAND_ROW4_LEFT	STAND_ROW4_LEFT had <think>

#	Referring expression	Correct answer	Raw model output
#237	the Row 4 support bracket where the left bearing fits, next to the large left gear	STAND_ROW4_LEFT	STAND_ROW4_LEFT had <think>
#238	the blue square-row left gear stand supporting the gear shaft	STAND_ROW4_LEFT	STAND_ROW4_LEFT had <think>
#239	the left Row 4 support that holds a bearing and supports the gear shaft	STAND_ROW4_LEFT	STAND_ROW4_LEFT had <think>
#241	blue left stand	STAND_ROW4_LEFT	STAND_ROW4_LEFT had <think>
#242	Row 4 left bracket	STAND_ROW4_LEFT	STAND_ROW4_LEFT had <think>
#243	left gear stand of Row 4	STAND_ROW4_LEFT	STAND_ROW4_LEFT had <think>
#244	blue left gear stand in the square row	STAND_ROW4_LEFT	STAND_ROW4_LEFT had <think>
#245	left bracket of the blue square row	STAND_ROW4_LEFT	STAND_ROW4_LEFT had <think>
#246	the Row 4 right gear stand	STAND_ROW4_RIGHT	STAND_ROW4_RIGHT had <think>
#247	the upright blue right stand with a circular bearing opening	STAND_ROW4_RIGHT	STAND_ROW4_RIGHT had <think>
#248	the right bearing support bracket in Row 4	STAND_ROW4_RIGHT	STAND_ROW4_RIGHT had <think>
#249	the Row 4 right stand where the bearing fits, at the right end of the blue gear rod	STAND_ROW4_RIGHT	STAND_ROW4_RIGHT had <think>
#250	the blue square-row right gear stand supporting the gear shaft	STAND_ROW4_RIGHT	STAND_ROW4_RIGHT had <think>
#251	the right Row 4 support that holds a bearing and supports the gear shaft	STAND_ROW4_RIGHT	STAND_ROW4_RIGHT had <think>
#253	blue right stand	STAND_ROW4_RIGHT	STAND_ROW4_RIGHT had <think>
#254	Row 4 right bracket	STAND_ROW4_RIGHT	STAND_ROW4_RIGHT had <think>
#255	right gear stand of Row 4	STAND_ROW4_RIGHT	STAND_ROW4_RIGHT had <think>
#256	blue right gear stand in the square row	STAND_ROW4_RIGHT	STAND_ROW4_RIGHT had <think>
#258	the Row 4 left spur gear	GEAR_ROW4_LEFT	GEAR_ROW4_LEFT had <think>
#260	the left-side gear on the shaft in Row 4	GEAR_ROW4_LEFT	GEAR_ROW4_LEFT had <think>
#261	the Row 4 gear next to the left blue bracket, the only gear in Row 4	GEAR_ROW4_LEFT	GEAR_ROW4_LEFT had <think>
#262	the large blue spur gear on the left side of the square-marked shaft	GEAR_ROW4_LEFT	GEAR_ROW4_LEFT had <think>
#263	the Row 4 spur gear that transmits rotation along the gear train	GEAR_ROW4_LEFT	GEAR_ROW4_LEFT had <think>
#264	the large blue toothed wheel on the left side, beside the gear shaft	GEAR_ROW4_LEFT	GEAR_ROW4_LEFT had <think>
#265	gear of Row 4	GEAR_ROW4_LEFT	GEAR_ROW4_LEFT had <think>
#270	a Row 4 Phillips mounting screw	SCREW_ROW4	SCREW_ROW4 had <think>
#274	the silver Phillips screw for the blue Row 4 stand	SCREW_ROW4	SCREW_ROW4 had <think>
#275	the Row 4 fastener tightened with a Phillips screwdriver	SCREW_ROW4	SCREW_ROW4 had <think>
#277	Row 4 screw	SCREW_ROW4	SCREW_ROW4 had <think>
#278	Phillips screw	SCREW_ROW4	SCREW_ROW4 had <think>

#	Referring expression	Correct answer	Raw model output
#279	the wooden retaining dowel	PIN	PIN had <think>
#281	the small peg positioned crosswise through a gear-shaft hole	PIN	PIN had <think>
#282	the dowel that prevents a gear or crank from sliding along its shaft	PIN	PIN had <think>
#284	the wooden retainer inserted through a shaft hole to stop an attached part sliding	PIN	PIN had <think>
#286	wooden pin	PIN	PIN had <think>
#288	retaining pin	PIN	PIN had <think>

qwen3-4b (Qwen/Qwen3-4B-Instruct-2507 · lenient 85.4% / strict 85.4%)

1. Valid but wrong 40 — answered a real label from the list, but not the correct one

#	Referring expression	Correct answer	Model said
#46	the Row 1 support bracket where the left bearing fits, next to the large left gear	STAND_ROW1_LEFT	BEARING
#53	white left gear stand in the circle row	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#58	the stand immediately to the left of the crank handle	STAND_ROW1_RIGHT	STAND_ROW1_LEFT
#100	the Row 2 left gear stand	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#101	the upright red left stand with a circular bearing opening	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#102	the left bearing support bracket in the second row from Row 1	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#103	the Row 2 support bracket where the left bearing fits, next to the small left gear	STAND_ROW2_LEFT	BEARING
#104	the red triangle-row left gear stand supporting the gear shaft	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#106	the upright red bearing bracket on the left, beside the small left gear in the triangle-marked row	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#107	red left stand	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#108	Row 2 left bracket	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#110	red left gear stand in the triangle row	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#111	left bracket of the red triangle row	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#112	the Row 2 right gear stand	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#113	the upright red right stand with a circular bearing opening	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#114	the right bearing support bracket in the second row from Row 1	STAND_ROW2_RIGHT	BEARING
#116	the red triangle-row right gear stand supporting the gear shaft	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#118	the upright red bearing bracket on the right, beside the medium right gear in the triangle-marked row	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#119	red right stand	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#120	Row 2 right bracket	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#122	red right gear stand in the triangle row	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#123	right bracket of the red triangle row	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#133	smaller red gear in the triangle row	GEAR_ROW2_LEFT	GEAR_ROW2_RIGHT
#168	the upright green left stand with a circular bearing opening	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#169	the left bearing support bracket in Row 3	STAND_ROW3_LEFT	BEARING

#	Referring expression	Correct answer	Model said
#170	the Row 3 support bracket where the left bearing fits, next to the small left gear	STAND_ROW3_LEFT	BEARING
#171	the green hexagon-row left gear stand supporting the gear shaft	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#173	the upright green bearing bracket on the left, beside the small left gear in the hexagon-marked row	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#174	green left stand	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#175	Row 3 left bracket	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#177	green left gear stand in the hexagon row	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#178	left bracket of the green hexagon row	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#180	the upright green right stand with a circular bearing opening	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#183	the green hexagon-row right gear stand supporting the gear shaft	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#185	the upright green bearing bracket on the right, beside the medium right gear in the hexagon-marked row	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#186	green right stand	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#187	Row 3 right bracket	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#189	green right gear stand in the hexagon row	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#190	right bracket of the green hexagon row	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#236	the left bearing support bracket in Row 4	STAND_ROW4_LEFT	BEARING

2. Invalid 2 — could not be matched to any of the 26 allowed labels

#	Referring expression	Correct answer	Raw model output
#62	white right stand	STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT
#65	white right gear stand in the circle row	STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT

3. Row/side over-specification 0 — answered a specific identifier (e.g. BEARING_ROW2_LEFT) instead of the grouped label (BEARING/PIN/SCREW_ROW{n})

None.

4. Correct, but blocked by reasoning chain 0 — the model identified the right part, but the raw output still contains a <think> block or other wrapping, so strict grading (which compares the full raw text) can never match

None.

llama32-vision-11b (meta-llama/Llama-3.2-11B-Vision-Instruct · lenient 83.3% / strict 82.6%)

1. Valid but wrong 45 — answered a real label from the list, but not the correct one

#	Referring expression	Correct answer	Model said
#14	the bearing through which a gear shaft passes	BEARING	GEAR_ROW1_LEFT
#24	the crank attached at the right end of the white gear rod	CRANK_HANDLE_ROW1	GEAR_ROW1_LEFT
#27	the bent white crank connected to the end of the Row 1 shaft	CRANK_HANDLE_ROW1	GEAR_ROW1_LEFT
#28	white handle	CRANK_HANDLE_ROW1	GEAR_ROW1_LEFT
#47	the white circle-row left gear stand supporting the gear shaft	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#51	Row 1 left bracket	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#52	left gear stand of Row 1	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#53	white left gear stand in the circle row	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#54	left bracket of the white circle row	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#58	the stand immediately to the left of the crank handle	STAND_ROW1_RIGHT	STAND_ROW1_LEFT
#66	right bracket of the white circle row	STAND_ROW1_RIGHT	GEAR_ROW1_LEFT
#104	the red triangle-row left gear stand supporting the gear shaft	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#108	Row 2 left bracket	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#109	left gear stand of Row 2	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#110	red left gear stand in the triangle row	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#111	left bracket of the red triangle row	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#112	the Row 2 right gear stand	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#114	the right bearing support bracket in the second row from Row 1	STAND_ROW2_RIGHT	STAND_ROW2_LEFT
#116	the red triangle-row right gear stand supporting the gear shaft	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#120	Row 2 right bracket	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#121	right gear stand of Row 2	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#122	red right gear stand in the triangle row	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#123	right bracket of the red triangle row	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#141	larger gear of Row 2	GEAR_ROW2_RIGHT	GEAR_ROW2_LEFT
#142	red larger gear	GEAR_ROW2_RIGHT	GEAR_ROW2_LEFT
#144	larger red gear in the triangle row	GEAR_ROW2_RIGHT	GEAR_ROW2_LEFT
#150	the black T25 Torx screw for the red Row 2 stand	SCREW_ROW2	STAND_ROW2_LEFT
#157	the shaft crossing Row 3 between the two green brackets	GEAR_ROW3_LEFT	GEAR_ROW3_LEFT
#171	the green hexagon-row left gear stand supporting the gear shaft	STAND_ROW3_LEFT	GEAR_ROW3_LEFT

#	Referring expression	Correct answer	Model said
#177	green left gear stand in the hexagon row	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#181	the right bearing support bracket in Row 3	STAND_ROW3_RIGHT	STAND_ROW3_LEFT
#183	the green hexagon-row right gear stand supporting the gear shaft	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#189	green right gear stand in the hexagon row	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#190	right bracket of the green hexagon row	STAND_ROW3_RIGHT	GEAR_ROW3_LEFT
#203	the medium green spur wheel	GEAR_ROW3_RIGHT	GEAR_ROW3_LEFT
#208	larger gear of Row 3	GEAR_ROW3_RIGHT	GEAR_ROW3_LEFT
#209	green larger gear	GEAR_ROW3_RIGHT	GEAR_ROW3_LEFT
#211	larger green gear in the hexagon row	GEAR_ROW3_RIGHT	GEAR_ROW3_LEFT
#217	the black H3 hex-socket screw for the green Row 3 stand	SCREW_ROW3	STAND_ROW3_LEFT
#242	Row 4 left bracket	STAND_ROW4_LEFT	GEAR_ROW4_LEFT
#248	the right bearing support bracket in Row 4	STAND_ROW4_RIGHT	STAND_ROW4_LEFT
#251	the right Row 4 support that holds a bearing and supports the gear shaft	STAND_ROW4_RIGHT	STAND_ROW4_LEFT
#257	right bracket of the blue square row	STAND_ROW4_RIGHT	STAND_ROW4_LEFT
#271	a small silver screw with a recessed Phillips head	SCREW_ROW4	SCREW_ROW1
#287	brown pin	PIN	GEAR_ROW2_LEFT

2. Invalid 3 — could not be matched to any of the 26 allowed labels

#	Referring expression	Correct answer	Raw model output
#62	white right stand	STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT
#64	right gear stand of Row 1	STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT
#65	white right gear stand in the circle row	STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT

3. Row/side over-specification 2 — answered a specific identifier (e.g. BEARING_ROW2_LEFT) instead of the grouped label (BEARING/PIN/SCREW_ROW{n})

#	Referring expression	Correct answer	Raw model output
#82	a fastener used to secure a Row 1 bearing stand to the baseboard	SCREW_ROW1	SCREW_ROW1_RIGHT
#272	a screw at the base of a Row 4 gear stand	SCREW_ROW4	SCREW_ROW4_LEFT

4. Correct, but blocked by reasoning chain 0 — the model identified the right part, but the raw output still contains a <think> block or other wrapping, so strict grading (which compares the full raw text) can never match

None.

gemma4-e2b (google/gemma-4-E2B-it · lenient 71.9% / strict 70.1%)

1. Valid but wrong 70 — answered a real label from the list, but not the correct one

#	Referring expression	Correct answer	Model said
#42	circle-row white shaft	GEAR_ROD_ROW1	GEAR_ROW1_LEFT
#43	the Row 1 left gear stand	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#44	the upright white left stand with a circular bearing opening	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#45	the left bearing support bracket in Row 1	STAND_ROW1_LEFT	BEARING
#47	the white circle-row left gear stand supporting the gear shaft	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#49	the upright white bearing bracket on the left, beside the large left gear in the circle-marked row	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#50	white left stand	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#51	Row 1 left bracket	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#53	white left gear stand in the circle row	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#54	left bracket of the white circle row	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#56	the upright white stand with an overlapping bearing slot and through-hole	STAND_ROW1_RIGHT	STAND_ROW1_LEFT
#58	the stand immediately to the left of the crank handle	STAND_ROW1_RIGHT	STAND_ROW1_LEFT
#59	the white right bearing bracket whose bearing recess overlaps a hole through the stand	STAND_ROW1_RIGHT	BEARING
#61	the white crank-side gear stand with both a bearing slot and a penetrating hole	STAND_ROW1_RIGHT	STAND_ROW1_LEFT
#66	right bracket of the white circle row	STAND_ROW1_RIGHT	GEAR_ROW1_LEFT
#99	triangle-row red shaft	GEAR_ROD_ROW2	GEAR_ROW2_RIGHT
#100	the Row 2 left gear stand	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#101	the upright red left stand with a circular bearing opening	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#102	the left bearing support bracket in the second row from Row 1	STAND_ROW2_LEFT	BEARING
#104	the red triangle-row left gear stand supporting the gear shaft	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#106	the upright red bearing bracket on the left, beside the small left gear in the triangle-marked row	STAND_ROW2_LEFT	BEARING
#107	red left stand	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#108	Row 2 left bracket	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#109	left gear stand of Row 2	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#110	red left gear stand in the triangle row	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#111	left bracket of the red triangle row	STAND_ROW2_LEFT	GEAR_ROW2_LEFT

#	Referring expression	Correct answer	Model said
#112	the Row 2 right gear stand	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#113	the upright red right stand with a circular bearing opening	STAND_ROW2_RIGHT	BEARING
#114	the right bearing support bracket in the second row from Row 1	STAND_ROW2_RIGHT	BEARING
#115	the Row 2 support bracket where the right bearing fits, next to the medium right gear	STAND_ROW2_RIGHT	BEARING
#116	the red triangle-row right gear stand supporting the gear shaft	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#117	the right Row 2 support that holds a bearing and supports the gear shaft	STAND_ROW2_RIGHT	BEARING
#118	the upright red bearing bracket on the right, beside the medium right gear in the triangle-marked row	STAND_ROW2_RIGHT	BEARING
#119	red right stand	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#120	Row 2 right bracket	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#122	red right gear stand in the triangle row	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#123	right bracket of the red triangle row	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#125	the small red spur wheel	GEAR_ROW2_LEFT	GEAR_ROW2_RIGHT
#133	smaller red gear in the triangle row	GEAR_ROW2_LEFT	GEAR_ROW2_RIGHT
#166	hexagon-row green shaft	GEAR_ROW3_RIGHT	GEAR_ROW3_RIGHT
#167	the Row 3 left gear stand	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#168	the upright green left stand with a circular bearing opening	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#169	the left bearing support bracket in Row 3	STAND_ROW3_LEFT	BEARING
#171	the green hexagon-row left gear stand supporting the gear shaft	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#173	the upright green bearing bracket on the left, beside the small left gear in the hexagon-marked row	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#174	green left stand	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#175	Row 3 left bracket	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#177	green left gear stand in the hexagon row	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#178	left bracket of the green hexagon row	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#180	the upright green right stand with a circular bearing opening	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#181	the right bearing support bracket in Row 3	STAND_ROW3_RIGHT	BEARING
#182	the Row 3 support bracket where the right bearing fits, next to the medium right gear	STAND_ROW3_RIGHT	BEARING
#183	the green hexagon-row right gear stand supporting the gear shaft	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#184	the right Row 3 support that holds a bearing and supports the gear shaft	STAND_ROW3_RIGHT	BEARING

#	Referring expression	Correct answer	Model said
#185	the upright green bearing bracket on the right, beside the medium right gear in the hexagon-marked row	STAND_ROW3_RIGHT	BEARING
#186	green right stand	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#187	Row 3 right bracket	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#189	green right gear stand in the hexagon row	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#190	right bracket of the green hexagon row	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#208	larger gear of Row 3	GEAR_ROW3_RIGHT	GEAR_ROW3_LEFT
#233	square-row blue shaft	GEAR_ROW4_RIGHT	GEAR_ROW4_LEFT
#235	the upright blue left stand with a circular bearing opening	STAND_ROW4_LEFT	GEAR_ROW4_LEFT
#236	the left bearing support bracket in Row 4	STAND_ROW4_LEFT	BEARING
#238	the blue square-row left gear stand supporting the gear shaft	STAND_ROW4_LEFT	GEAR_ROW4_LEFT
#240	the upright blue bearing bracket on the left, beside the large left gear in the square-marked row	STAND_ROW4_LEFT	GEAR_ROW4_LEFT
#241	blue left stand	STAND_ROW4_LEFT	GEAR_ROW4_LEFT
#242	Row 4 left bracket	STAND_ROW4_LEFT	GEAR_ROW4_LEFT
#244	blue left gear stand in the square row	STAND_ROW4_LEFT	GEAR_ROW4_LEFT
#245	left bracket of the blue square row	STAND_ROW4_LEFT	GEAR_ROW4_LEFT
#248	the right bearing support bracket in Row 4	STAND_ROW4_RIGHT	BEARING

2. Invalid 11 — could not be matched to any of the 26 allowed labels

#	Referring expression	Correct answer	Raw model output
#57	the right-hand support in circle-marked Row 1	STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT
#62	white right stand	STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT
#63	Row 1 right bracket	STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT
#65	white right gear stand in the circle row	STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT
#247	the upright blue right stand with a circular bearing opening	STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT
#250	the blue square-row right gear stand supporting the gear shaft	STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT
#252	the upright blue bearing bracket on the right, beside the crank-side end of the shaft in the square-marked row	STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT
#253	blue right stand	STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT
#254	Row 4 right bracket	STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT
#256	blue right gear stand in the square row	STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT
#257	right bracket of the blue square row	STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT

3. Row/side over-specification 2 — answered a specific identifier (e.g. BEARING_ROW2_LEFT) instead of the grouped label (BEARING/PIN/SCREW_ROW{n})

#	Referring expression	Correct answer	Raw model output
#81	a screw at the base of a Row 1 gear stand	SCREW_ROW1	SCREW_ROW1_LEFT
#274	the silver Phillips screw for the blue Row 4 stand	SCREW_ROW4	SCREW_ROW4_LEFT

4. Correct, but blocked by reasoning chain 3 — the model identified the right part, but the raw output still contains a <think> block or other wrapping, so strict grading (which compares the full raw text) can never match

#	Referring expression	Correct answer	Raw model output
#7	the broad black rectangular base beneath the white, red, green, and blue mechanisms	BASE_BOARD	<code>``text</code> BASE_BOARD <code>``</code>
#126	the left-side gear on the shaft in the second row from Row 1	GEAR_ROW2_LEFT	<code>``text</code> GEAR_ROW2_LEFT <code>``</code>
#137	the right-side gear on the shaft in the second row from Row 1	GEAR_ROW2_RIGHT	<code>``text</code> GEAR_ROW2_RIGHT <code>``</code>

gemma3-4b (google/gemma-3-4b-it · lenient 62.8% / strict 62.2%)

1. Valid but wrong 97 — answered a real label from the list, but not the correct one

#	Referring expression	Correct answer	Model said
#20	gear-stand bearing	BEARING	STAND_ROW1_LEFT
#32	the long white cylindrical shaft with two pin holes	GEAR_ROD_ROW1	GEAR_ROD_ROW4
#35	the long white circle-row gear shaft seated in both bearing brackets	GEAR_ROD_ROW1	GEAR_ROD_ROW4
#42	circle-row white shaft	GEAR_ROD_ROW1	GEAR_ROW1_LEFT
#43	the Row 1 left gear stand	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#44	the upright white left stand with a circular bearing opening	STAND_ROW1_LEFT	BEARING
#47	the white circle-row left gear stand supporting the gear shaft	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#48	the left Row 1 support that holds a bearing and supports the gear shaft	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#49	the upright white bearing bracket on the left, beside the large left gear in the circle-marked row	STAND_ROW1_LEFT	GEAR_ROW4_LEFT
#50	white left stand	STAND_ROW1_LEFT	GEAR_ROW4_LEFT
#51	Row 1 left bracket	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#52	left gear stand of Row 1	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#53	white left gear stand in the circle row	STAND_ROW1_LEFT	GEAR_ROW2_LEFT
#54	left bracket of the white circle row	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#56	the upright white stand with an overlapping bearing slot and through-hole	STAND_ROW1_RIGHT	STAND_ROW4_RIGHT
#58	the stand immediately to the left of the crank handle	STAND_ROW1_RIGHT	STAND_ROW1_LEFT
#59	the white right bearing bracket whose bearing recess overlaps a hole through the stand	STAND_ROW1_RIGHT	BEARING
#61	the white crank-side gear stand with both a bearing slot and a penetrating hole	STAND_ROW1_RIGHT	STAND_ROW4_RIGHT
#62	white right stand	STAND_ROW1_RIGHT	GEAR_ROW4_LEFT
#63	Row 1 right bracket	STAND_ROW1_RIGHT	BEARING
#65	white right gear stand in the circle row	STAND_ROW1_RIGHT	GEAR_ROW4_LEFT
#66	right bracket of the white circle row	STAND_ROW1_RIGHT	STAND_ROW4_RIGHT
#68	the large white spur wheel	GEAR_ROW1_LEFT	GEAR_ROW4_LEFT
#71	the large white spur gear on the left side of the circle-marked shaft	GEAR_ROW1_LEFT	GEAR_ROW2_LEFT
#73	the large white toothed wheel on the left side, beside the gear shaft	GEAR_ROW1_LEFT	GEAR_ROW4_LEFT
#76	Row 1 gear	GEAR_ROW1_LEFT	GEAR_ROD_ROW1

#	Referring expression	Correct answer	Model said
#80	a small silver screw with a recessed H5 hex-socket head	SCREW_ROW1	SCREW_ROW2
#85	the small silver screw located at the foot of a white support bracket	SCREW_ROW1	SCREW_ROW4
#99	triangle-row red shaft	GEAR_ROD_ROW2	GEAR_ROW2_LEFT
#100	the Row 2 left gear stand	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#101	the upright red left stand with a circular bearing opening	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#102	the left bearing support bracket in the second row from Row 1	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#103	the Row 2 support bracket where the left bearing fits, next to the small left gear	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#104	the red triangle-row left gear stand supporting the gear shaft	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#105	the left Row 2 support that holds a bearing and supports the gear shaft	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#106	the upright red bearing bracket on the left, beside the small left gear in the triangle-marked row	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#107	red left stand	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#108	Row 2 left bracket	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#109	left gear stand of Row 2	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#110	red left gear stand in the triangle row	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#111	left bracket of the red triangle row	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#112	the Row 2 right gear stand	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#113	the upright red right stand with a circular bearing opening	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#114	the right bearing support bracket in the second row from Row 1	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#116	the red triangle-row right gear stand supporting the gear shaft	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#117	the right Row 2 support that holds a bearing and supports the gear shaft	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#118	the upright red bearing bracket on the right, beside the medium right gear in the triangle-marked row	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#119	red right stand	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#120	Row 2 right bracket	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#121	right gear stand of Row 2	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#122	red right gear stand in the triangle row	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#123	right bracket of the red triangle row	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#133	smaller red gear in the triangle row	GEAR_ROW2_LEFT	GEAR_ROW2_RIGHT
#166	hexagon-row green shaft	GEAR_ROD_ROW3	GEAR_ROW3_LEFT
#167	the Row 3 left gear stand	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#168	the upright green left stand with a circular bearing opening	STAND_ROW3_LEFT	GEAR_ROW3_LEFT

#	Referring expression	Correct answer	Model said
#170	the Row 3 support bracket where the left bearing fits, next to the small left gear	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#171	the green hexagon-row left gear stand supporting the gear shaft	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#172	the left Row 3 support that holds a bearing and supports the gear shaft	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#173	the upright green bearing bracket on the left, beside the small left gear in the hexagon-marked row	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#174	green left stand	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#175	Row 3 left bracket	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#176	left gear stand of Row 3	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#177	green left gear stand in the hexagon row	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#178	left bracket of the green hexagon row	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#179	the Row 3 right gear stand	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#180	the upright green right stand with a circular bearing opening	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#183	the green hexagon-row right gear stand supporting the gear shaft	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#184	the right Row 3 support that holds a bearing and supports the gear shaft	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#185	the upright green bearing bracket on the right, beside the medium right gear in the hexagon-marked row	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#186	green right stand	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#187	Row 3 right bracket	STAND_ROW3_RIGHT	GEAR_ROD_ROW3
#188	right gear stand of Row 3	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#189	green right gear stand in the hexagon row	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#193	the left-side gear on the shaft in Row 3	GEAR_ROW3_LEFT	GEAR_ROD_ROW3
#194	the Row 3 gear next to the left green bracket, smaller than the right gear	GEAR_ROW3_LEFT	GEAR_ROD_ROW3
#204	the right-side gear on the shaft in Row 3	GEAR_ROW3_RIGHT	GEAR_ROD_ROW3
#205	the Row 3 gear next to the right green bracket, larger than the left gear	GEAR_ROW3_RIGHT	GEAR_ROD_ROW3
#208	larger gear of Row 3	GEAR_ROW3_RIGHT	GEAR_ROD_ROW3
#209	green larger gear	GEAR_ROW3_RIGHT	GEAR_ROW3_LEFT
#210	right gear of Row 3	GEAR_ROW3_RIGHT	GEAR_ROD_ROW3
#233	square-row blue shaft	GEAR_ROD_ROW4	GEAR_ROW4_LEFT
#234	the Row 4 left gear stand	STAND_ROW4_LEFT	GEAR_ROW4_LEFT
#235	the upright blue left stand with a circular bearing opening	STAND_ROW4_LEFT	BEARING
#238	the blue square-row left gear stand supporting the gear shaft	STAND_ROW4_LEFT	GEAR_ROW4_LEFT

#	Referring expression	Correct answer	Model said
#239	the left Row 4 support that holds a bearing and supports the gear shaft	STAND_ROW4_LEFT	GEAR_ROW4_LEFT
#240	the upright blue bearing bracket on the left, beside the large left gear in the square-marked row	STAND_ROW4_LEFT	GEAR_ROW4_LEFT
#241	blue left stand	STAND_ROW4_LEFT	GEAR_ROW4_LEFT
#242	Row 4 left bracket	STAND_ROW4_LEFT	BEARING
#243	left gear stand of Row 4	STAND_ROW4_LEFT	GEAR_ROW4_LEFT
#244	blue left gear stand in the square row	STAND_ROW4_LEFT	GEAR_ROW4_LEFT
#245	left bracket of the blue square row	STAND_ROW4_LEFT	GEAR_ROW4_LEFT
#247	the upright blue right stand with a circular bearing opening	STAND_ROW4_RIGHT	BEARING
#249	the Row 4 right stand where the bearing fits, at the right end of the blue gear rod	STAND_ROW4_RIGHT	BEARING
#254	Row 4 right bracket	STAND_ROW4_RIGHT	SCREW_ROW4
#260	the left-side gear on the shaft in Row 4	GEAR_ROW4_LEFT	GEAR_ROD_ROW4
#271	a small silver screw with a recessed Phillips head	SCREW_ROW4	SCREW_ROW2

2. Invalid 10 — could not be matched to any of the 26 allowed labels

#	Referring expression	Correct answer	Raw model output
#55	the Row 1 right gear stand	STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT
#57	the right-hand support in circle-marked Row 1	STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT
#64	right gear stand of Row 1	STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT
#246	the Row 4 right gear stand	STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT
#250	the blue square-row right gear stand supporting the gear shaft	STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT
#251	the right Row 4 support that holds a bearing and supports the gear shaft	STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT
#252	the upright blue bearing bracket on the right, beside the crank-side end of the shaft in the square-marked row	STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT
#253	blue right stand	STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT
#255	right gear stand of Row 4	STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT
#256	blue right gear stand in the square row	STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT

3. Row/side over-specification 2 — answered a specific identifier (e.g. BEARING_ROW2_LEFT) instead of the grouped label (BEARING/PIN/SCREW_ROW{n})

#	Referring expression	Correct answer	Raw model output
#18	black bearing	BEARING	BEARING_ROW2_LEFT
#285	the short brown dowel crossing the gear rod beside a gear or the crank	PIN	PIN_ROW2_LEFT

4. Correct, but blocked by reasoning chain  — the model identified the right part, but the raw output still contains a <think> block or other wrapping, so strict grading (which compares the full raw text) can never match

None.

qwen25-3b (Qwen/Qwen2.5-3B-Instruct · lenient 62.2% / strict 46.9%)

1. Valid but wrong 95 — answered a real label from the list, but not the correct one

#	Referring expression	Correct answer	Model said
#2	the large flat black rectangular plate	BASE_BOARD	GEAR_ROW4_LEFT
#4	the plate to which all eight bearing support brackets are screwed	BASE_BOARD	STAND_ROW1_LEFT
#5	the large black rectangular mounting plate below every gear shaft and stand	BASE_BOARD	GEAR_ROW2_RIGHT
#6	the rigid plate that supports and locates every gearbox row	BASE_BOARD	STAND_ROW1_LEFT
#32	the long white cylindrical shaft with two pin holes	GEAR_ROD_ROW1	GEAR_ROD_ROW2
#34	the shaft that passes through both Row 1 bearings and carries one large gear	GEAR_ROD_ROW1	GEAR_ROW1_LEFT
#35	the long white circle-row gear shaft seated in both bearing brackets	GEAR_ROD_ROW1	GEAR_ROD_ROW2
#38	white gear rod	GEAR_ROD_ROW1	GEAR_ROD_ROW2
#41	white gear rod in the circle row	GEAR_ROD_ROW1	GEAR_ROD_ROW2
#42	circle-row white shaft	GEAR_ROD_ROW1	GEAR_ROW1_LEFT
#43	the Row 1 left gear stand	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#44	the upright white left stand with a circular bearing opening	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#45	the left bearing support bracket in Row 1	STAND_ROW1_LEFT	BEARING
#47	the white circle-row left gear stand supporting the gear shaft	STAND_ROW1_LEFT	GEAR_ROW2_LEFT
#49	the upright white bearing bracket on the left, beside the large left gear in the circle-marked row	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#51	Row 1 left bracket	STAND_ROW1_LEFT	BEARING
#52	left gear stand of Row 1	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#53	white left gear stand in the circle row	STAND_ROW1_LEFT	GEAR_ROW2_LEFT
#54	left bracket of the white circle row	STAND_ROW1_LEFT	GEAR_ROW2_LEFT
#56	the upright white stand with an overlapping bearing slot and through-hole	STAND_ROW1_RIGHT	STAND_ROW3_RIGHT
#58	the stand immediately to the left of the crank handle	STAND_ROW1_RIGHT	STAND_ROW1_LEFT
#59	the white right bearing bracket whose bearing recess overlaps a hole through the stand	STAND_ROW1_RIGHT	BEARING
#60	the right Row 1 support that holds the bearing and guides the gear shaft	STAND_ROW1_RIGHT	GEAR_ROD_ROW1
#65	white right gear stand in the circle row	STAND_ROW1_RIGHT	GEAR_ROW2_RIGHT
#66	right bracket of the white circle row	STAND_ROW1_RIGHT	GEAR_ROW2_RIGHT
#68	the large white spur wheel	GEAR_ROW1_LEFT	GEAR_ROW4_LEFT

#	Referring expression	Correct answer	Model said
#73	the large white toothed wheel on the left side, beside the gear shaft	GEAR_ROW1_LEFT	GEAR_ROW4_LEFT
#75	white gear	GEAR_ROW1_LEFT	GEAR_ROW2_LEFT
#77	white gear in the circle row	GEAR_ROW1_LEFT	GEAR_ROW2_LEFT
#80	a small silver screw with a recessed H5 hex-socket head	SCREW_ROW1	SCREW_ROW2
#85	the small silver screw located at the foot of a white support bracket	SCREW_ROW1	SCREW_ROW3
#99	triangle-row red shaft	GEAR_ROD_ROW2	GEAR_ROW2_RIGHT
#100	the Row 2 left gear stand	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#101	the upright red left stand with a circular bearing opening	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#102	the left bearing support bracket in the second row from Row 1	STAND_ROW2_LEFT	BEARING
#104	the red triangle-row left gear stand supporting the gear shaft	STAND_ROW2_LEFT	GEAR_ROD_ROW2
#106	the upright red bearing bracket on the left, beside the small left gear in the triangle-marked row	STAND_ROW2_LEFT	BEARING
#108	Row 2 left bracket	STAND_ROW2_LEFT	BEARING
#109	left gear stand of Row 2	STAND_ROW2_LEFT	GEAR_ROD_ROW2
#110	red left gear stand in the triangle row	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#111	left bracket of the red triangle row	STAND_ROW2_LEFT	GEAR_ROW2_LEFT
#112	the Row 2 right gear stand	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#113	the upright red right stand with a circular bearing opening	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#114	the right bearing support bracket in the second row from Row 1	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#116	the red triangle-row right gear stand supporting the gear shaft	STAND_ROW2_RIGHT	GEAR_ROD_ROW2
#117	the right Row 2 support that holds a bearing and supports the gear shaft	STAND_ROW2_RIGHT	GEAR_ROD_ROW2
#118	the upright red bearing bracket on the right, beside the medium right gear in the triangle-marked row	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#121	right gear stand of Row 2	STAND_ROW2_RIGHT	GEAR_ROD_ROW2
#122	red right gear stand in the triangle row	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#123	right bracket of the red triangle row	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#157	the shaft crossing Row 3 between the two green brackets	GEAR_ROD_ROW3	GEAR_ROW3_RIGHT
#159	the long green hexagon-row gear shaft seated in both bearing brackets	GEAR_ROD_ROW3	GEAR_ROW3_LEFT
#163	Row 3 shaft	GEAR_ROD_ROW3	GEAR_ROW3_RIGHT
#166	hexagon-row green shaft	GEAR_ROD_ROW3	GEAR_ROW3_LEFT
#167	the Row 3 left gear stand	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#168	the upright green left stand with a circular bearing opening	STAND_ROW3_LEFT	GEAR_ROW3_LEFT

#	Referring expression	Correct answer	Model said
#169	the left bearing support bracket in Row 3	STAND_ROW3_LEFT	BEARING
#170	the Row 3 support bracket where the left bearing fits, next to the small left gear	STAND_ROW3_LEFT	BEARING
#171	the green hexagon-row left gear stand supporting the gear shaft	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#173	the upright green bearing bracket on the left, beside the small left gear in the hexagon-marked row	STAND_ROW3_LEFT	BEARING
#175	Row 3 left bracket	STAND_ROW3_LEFT	BEARING
#176	left gear stand of Row 3	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#177	green left gear stand in the hexagon row	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#178	left bracket of the green hexagon row	STAND_ROW3_LEFT	GEAR_ROW3_LEFT
#179	the Row 3 right gear stand	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#180	the upright green right stand with a circular bearing opening	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#181	the right bearing support bracket in Row 3	STAND_ROW3_RIGHT	BEARING
#183	the green hexagon-row right gear stand supporting the gear shaft	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#184	the right Row 3 support that holds a bearing and supports the gear shaft	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#185	the upright green bearing bracket on the right, beside the medium right gear in the hexagon-marked row	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#188	right gear stand of Row 3	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#189	green right gear stand in the hexagon row	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#190	right bracket of the green hexagon row	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#225	the shaft that passes through both Row 4 bearings and carries one large gear	GEAR_ROD_ROW4	GEAR_ROW4_LEFT
#230	Row 4 shaft	GEAR_ROD_ROW4	GEAR_ROW4_LEFT
#233	square-row blue shaft	GEAR_ROD_ROW4	GEAR_ROW4_LEFT
#234	the Row 4 left gear stand	STAND_ROW4_LEFT	GEAR_ROW4_LEFT
#235	the upright blue left stand with a circular bearing opening	STAND_ROW4_LEFT	GEAR_ROW4_LEFT
#236	the left bearing support bracket in Row 4	STAND_ROW4_LEFT	BEARING
#238	the blue square-row left gear stand supporting the gear shaft	STAND_ROW4_LEFT	GEAR_ROW4_LEFT
#239	the left Row 4 support that holds a bearing and supports the gear shaft	STAND_ROW4_LEFT	GEAR_ROD_ROW4
#240	the upright blue bearing bracket on the left, beside the large left gear in the square-marked row	STAND_ROW4_LEFT	BEARING
#242	Row 4 left bracket	STAND_ROW4_LEFT	BEARING

#	Referring expression	Correct answer	Model said
#243	left gear stand of Row 4	STAND_ROW4_LEFT	GEAR_ROW4_LEFT
#244	blue left gear stand in the square row	STAND_ROW4_LEFT	GEAR_ROW4_LEFT
#245	left bracket of the blue square row	STAND_ROW4_LEFT	GEAR_ROW4_LEFT
#247	the upright blue right stand with a circular bearing opening	STAND_ROW4_RIGHT	GEAR_ROW4_LEFT
#248	the right bearing support bracket in Row 4	STAND_ROW4_RIGHT	BEARING
#249	the Row 4 right stand where the bearing fits, at the right end of the blue gear rod	STAND_ROW4_RIGHT	GEAR_ROD_ROW4
#250	the blue square-row right gear stand supporting the gear shaft	STAND_ROW4_RIGHT	GEAR_ROD_ROW4
#251	the right Row 4 support that holds a bearing and supports the gear shaft	STAND_ROW4_RIGHT	GEAR_ROD_ROW4
#252	the upright blue bearing bracket on the right, beside the crank-side end of the shaft in the square-marked row	STAND_ROW4_RIGHT	GEAR_ROW4_LEFT
#255	right gear stand of Row 4	STAND_ROW4_RIGHT	GEAR_ROW4_LEFT
#271	a small silver screw with a recessed Phillips head	SCREW_ROW4	SCREW_ROW2
#278	Phillips screw	SCREW_ROW4	SCREW_ROW1

2. Invalid 14 — could not be matched to any of the 26 allowed labels

#	Referring expression	Correct answer	Raw model output
#22	the white bent lever with a rounded grip	CRANK_HANDLE_ROW1	INVALID_LABEL
#55	the Row 1 right gear stand	STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT
#57	the right-hand support in circle-marked Row 1	STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT
#62	white right stand	STAND_ROW1_RIGHT	GEAR_STAND_ROW2_RIGHT
#64	right gear stand of Row 1	STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT
#119	red right stand	STAND_ROW2_RIGHT	GEAR_STAND_ROW2_RIGHT
#174	green left stand	STAND_ROW3_LEFT	GEAR_STAND_ROW3_LEFT
#186	green right stand	STAND_ROW3_RIGHT	GEAR_STAND_ROW3_RIGHT
#241	blue left stand	STAND_ROW4_LEFT	GEAR_STAND_ROW4_LEFT
#246	the Row 4 right gear stand	STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT
#253	blue right stand	STAND_ROW4_RIGHT	GEAR_STAND_ROW4_RIGHT
#254	Row 4 right bracket	STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT
#256	blue right gear stand in the square row	STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT
#257	right bracket of the blue square row	STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT

3. Row/side over-specification 44 — answered a specific identifier (e.g. BEARING_ROW2_LEFT) instead of the grouped label (BEARING/PIN/SCREW_ROW{n})

#	Referring expression	Correct answer	Raw model output
#14	the bearing through which a gear shaft passes	BEARING	BEARING_Row2_LEFT

#	Referring expression	Correct answer	Raw model output
#15	the small black ring-shaped bearing that fits into each colored stand	BEARING	BEARING_RowX_LEFT
#16	the black bearing that supports a rotating gear shaft inside a stand	BEARING	BEARING_ROW2_LEFT
#17	the small black ring seated in the round opening of a colored support bracket	BEARING	BEARING_Row2_LEFT
#18	black bearing	BEARING	BEARING_Row2_LEFT
#20	gear-stand bearing	BEARING	BEARING_STAND_ROW2_LEFT
#79	a Row 1 H5 hex-socket mounting screw	SCREW_ROW1	SCREW_ROW1_RIGHT
#81	a screw at the base of a Row 1 gear stand	SCREW_ROW1	SCREW_ROW1_LEFT
#82	a fastener used to secure a Row 1 bearing stand to the baseboard	SCREW_ROW1	SCREW_ROW1_LEFT
#83	the silver H5 hex-socket screw for the white Row 1 stand	SCREW_ROW1	SCREW_ROW1_LEFT
#84	the Row 1 fastener tightened with a 5 mm hex key	SCREW_ROW1	SCREW_ROW1_RIGHT
#87	H5 screw	SCREW_ROW1	SCREW_ROW1_RIGHT
#146	a Row 2 T25 Torx mounting screw	SCREW_ROW2	SCREW_ROW2_RIGHT
#147	a small black screw with a recessed T25 Torx head	SCREW_ROW2	SCREW_ROW2_RIGHT
#148	a screw at the base of a Row 2 gear stand	SCREW_ROW2	SCREW_ROW2_LEFT
#149	a fastener used to secure a Row 2 bearing stand to the baseboard	SCREW_ROW2	SCREW_ROW2_RIGHT
#150	the black T25 Torx screw for the red Row 2 stand	SCREW_ROW2	SCREW_ROW2_RIGHT
#151	the Row 2 fastener tightened with a T25 driver	SCREW_ROW2	SCREW_ROW2_RIGHT
#152	the small black screw located at the foot of a red support bracket	SCREW_ROW2	SCREW_ROW2_RIGHT
#154	T25 screw	SCREW_ROW2	SCREW_ROW2_RIGHT
#213	a Row 3 H3 hex-socket mounting screw	SCREW_ROW3	SCREW_ROW3_RIGHT
#214	a small black screw with a recessed H3 hex-socket head	SCREW_ROW3	SCREW_ROW3_RIGHT
#215	a screw at the base of a Row 3 gear stand	SCREW_ROW3	SCREW_ROW3_LEFT
#216	a fastener used to secure a Row 3 bearing stand to the baseboard	SCREW_ROW3	SCREW_ROW3_RIGHT
#217	the black H3 hex-socket screw for the green Row 3 stand	SCREW_ROW3	SCREW_ROW3_LEFT
#218	the Row 3 fastener tightened with a 3 mm hex key	SCREW_ROW3	SCREW_ROW3_RIGHT
#219	the small black screw located at the foot of a green support bracket	SCREW_ROW3	SCREW_ROW3_RIGHT
#220	Row 3 screw	SCREW_ROW3	SCREW_ROW3_RIGHT

#	Referring expression	Correct answer	Raw model output
#221	H3 screw	SCREW_ROW3	SCREW_ROW3_RIGHT
#270	a Row 4 Phillips mounting screw	SCREW_ROW4	SCREW_ROW4_LEFT
#272	a screw at the base of a Row 4 gear stand	SCREW_ROW4	SCREW_ROW4_LEFT
#273	a fastener used to secure a Row 4 bearing stand to the baseboard	SCREW_ROW4	SCREW_ROW4_LEFT
#274	the silver Phillips screw for the blue Row 4 stand	SCREW_ROW4	SCREW_ROW4_LEFT
#275	the Row 4 fastener tightened with a Phillips screwdriver	SCREW_ROW4	SCREW_ROW4_RIGHT
#276	the small silver screw located at the foot of a blue support bracket	SCREW_ROW4	SCREW_ROW4_LEFT
#277	Row 4 screw	SCREW_ROW4	SCREW_ROW4_RIGHT
#279	the wooden retaining dowel	PIN	PIN_RowX
#280	the short thin brown cylindrical peg	PIN	PIN_ROW2_LEFT
#281	the small peg positioned crosswise through a gear-shaft hole	PIN	PIN_Row2_LEFT
#283	the small brown cylindrical retaining pin fitted beside a gear	PIN	PIN_ROW2_LEFT
#285	the short brown dowel crossing the gear rod beside a gear or the crank	PIN	PIN_ROW1_RIGHT
#286	wooden pin	PIN	PIN_Row2_Left
#287	brown pin	PIN	PIN_Row2_RIGHT
#288	retaining pin	PIN	PIN_Row2_RIGHT

4. Correct, but blocked by reasoning chain 0 — the model identified the right part, but the raw output still contains a <think> block or other wrapping, so strict grading (which compares the full raw text) can never match

None.

qwen25-vl-3b (Qwen/Qwen2.5-VL-3B-Instruct · lenient 52.8% / strict 35.8%)

1. Valid but wrong 50 — answered a real label from the list, but not the correct one

#	Referring expression	Correct answer	Model said
#39	Row 1 shaft	GEAR_ROD_ROW1	GEAR_ROW1_LEFT
#42	circle-row white shaft	GEAR_ROD_ROW1	GEAR_ROW1_LEFT
#44	the upright white left stand with a circular bearing opening	STAND_ROW1_LEFT	BEARING
#49	the upright white bearing bracket on the left, beside the large left gear in the circle-marked row	STAND_ROW1_LEFT	BEARING
#51	Row 1 left bracket	STAND_ROW1_LEFT	BEARING
#54	left bracket of the white circle row	STAND_ROW1_LEFT	GEAR_ROW1_LEFT
#58	the stand immediately to the left of the crank handle	STAND_ROW1_RIGHT	STAND_ROW1_LEFT
#62	white right stand	STAND_ROW1_RIGHT	GEAR_ROW2_RIGHT
#65	white right gear stand in the circle row	STAND_ROW1_RIGHT	GEAR_ROW2_RIGHT
#66	right bracket of the white circle row	STAND_ROW1_RIGHT	GEAR_ROW2_RIGHT
#99	triangle-row red shaft	GEAR_ROD_ROW2	GEAR_ROW2_RIGHT
#100	the Row 2 left gear stand	STAND_ROW2_LEFT	BEARING
#101	the upright red left stand with a circular bearing opening	STAND_ROW2_LEFT	BEARING
#102	the left bearing support bracket in the second row from Row 1	STAND_ROW2_LEFT	BEARING
#103	the Row 2 support bracket where the left bearing fits, next to the small left gear	STAND_ROW2_LEFT	BEARING
#105	the left Row 2 support that holds a bearing and supports the gear shaft	STAND_ROW2_LEFT	BEARING
#106	the upright red bearing bracket on the left, beside the small left gear in the triangle-marked row	STAND_ROW2_LEFT	BEARING
#107	red left stand	STAND_ROW2_LEFT	BEARING
#108	Row 2 left bracket	STAND_ROW2_LEFT	BEARING
#109	left gear stand of Row 2	STAND_ROW2_LEFT	BEARING
#110	red left gear stand in the triangle row	STAND_ROW2_LEFT	BEARING
#111	left bracket of the red triangle row	STAND_ROW2_LEFT	BEARING
#114	the right bearing support bracket in the second row from Row 1	STAND_ROW2_RIGHT	BEARING
#115	the Row 2 support bracket where the right bearing fits, next to the medium right gear	STAND_ROW2_RIGHT	BEARING
#117	the right Row 2 support that holds a bearing and supports the gear shaft	STAND_ROW2_RIGHT	BEARING
#119	red right stand	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#120	Row 2 right bracket	STAND_ROW2_RIGHT	BEARING

#	Referring expression	Correct answer	Model said
#122	red right gear stand in the triangle row	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#123	right bracket of the red triangle row	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT
#133	smaller red gear in the triangle row	GEAR_ROW2_LEFT	GEAR_ROW2_RIGHT
#141	larger gear of Row 2	GEAR_ROW2_RIGHT	GEAR_ROW2_LEFT
#157	the shaft crossing Row 3 between the two green brackets	GEAR_ROD_ROW3	GEAR_ROW3_LEFT
#159	the long green hexagon-row gear shaft seated in both bearing brackets	GEAR_ROD_ROW3	GEAR_ROW3_LEFT
#163	Row 3 shaft	GEAR_ROD_ROW3	GEAR_ROW3_LEFT
#166	hexagon-row green shaft	GEAR_ROD_ROW3	GEAR_ROW3_LEFT
#171	the green hexagon-row left gear stand supporting the gear shaft	STAND_ROW3_LEFT	BEARING
#173	the upright green bearing bracket on the left, beside the small left gear in the hexagon-marked row	STAND_ROW3_LEFT	BEARING
#174	green left stand	STAND_ROW3_LEFT	BEARING
#175	Row 3 left bracket	STAND_ROW3_LEFT	BEARING
#178	left bracket of the green hexagon row	STAND_ROW3_LEFT	BEARING
#182	the Row 3 support bracket where the right bearing fits, next to the medium right gear	STAND_ROW3_RIGHT	BEARING
#184	the right Row 3 support that holds a bearing and supports the gear shaft	STAND_ROW3_RIGHT	BEARING
#190	right bracket of the green hexagon row	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT
#208	larger gear of Row 3	GEAR_ROW3_RIGHT	GEAR_ROW3_LEFT
#209	green larger gear	GEAR_ROW3_RIGHT	GEAR_ROW3_LEFT
#211	larger green gear in the hexagon row	GEAR_ROW3_RIGHT	GEAR_ROW3_LEFT
#233	square-row blue shaft	GEAR_ROD_ROW4	GEAR_ROW4_LEFT
#245	left bracket of the blue square row	STAND_ROW4_LEFT	GEAR_ROW4_LEFT
#271	a small silver screw with a recessed Phillips head	SCREW_ROW4	SCREW_ROW1
#278	Phillips screw	SCREW_ROW4	SCREW_ROW1

2. Invalid 86 — could not be matched to any of the 26 allowed labels

#	Referring expression	Correct answer	Raw model output
#20	gear-stand bearing	BEARING	STAND_ROW1_RIGHT_ASSEMBLY
#23	the handle immediately to the right of the right white gear stand	CRANK_HANDLE_ROW1	CRANK_HANDLE_ROW1_ASSEMBLY
#24	the crank attached at the right end of the white gear rod	CRANK_HANDLE_ROW1	CRANK_HANDLE_ROW1_ASSEMBLY
#32	the long white cylindrical shaft with two pin holes	GEAR_ROD_ROW1	GEAR_ROD_ROW1_ASSEMBLY
#33	the shaft crossing Row 1 between the two white brackets	GEAR_ROD_ROW1	GEAR_ROD_ROW1_ASSEMBLY
#34	the shaft that passes through both Row 1 bearings and carries one large gear	GEAR_ROD_ROW1	GEAR_ROD_ROW1_ASSEMBLY

#	Referring expression	Correct answer	Raw model output
#35	the long white circle-row gear shaft seated in both bearing brackets	GEAR_ROD_ROW1	GEAR_ROD_ROW1_ASSEMBLY
#36	the Row 1 shaft that carries one large gear and transfers rotation between the two stands	GEAR_ROD_ROW1	GEAR_ROD_ROW1_ASSEMBLY
#37	the long white cylindrical rod with two retaining-pin holes spanning Row 1	GEAR_ROD_ROW1	GEAR_ROD_ROW1_ASSEMBLY
#46	the Row 1 support bracket where the left bearing fits, next to the large left gear	STAND_ROW1_LEFT	STAND_ROW1_RIGHT_ASSEMBLY
#47	the white circle-row left gear stand supporting the gear shaft	STAND_ROW1_LEFT	STAND_ROW1_LEFT_ASSEMBLY
#53	white left gear stand in the circle row	STAND_ROW1_LEFT	GEAR_ROW1_LEFT_ASSEMBLY
#55	the Row 1 right gear stand	STAND_ROW1_RIGHT	STAND_ROW1_RIGHT_ASSEMBLY
#56	the upright white stand with an overlapping bearing slot and through-hole	STAND_ROW1_RIGHT	STAND_ROW4_RIGHT_ASSEMBLY
#57	the right-hand support in circle-marked Row 1	STAND_ROW1_RIGHT	STAND_ROW1_RIGHT_ASSEMBLY
#59	the white right bearing bracket whose bearing recess overlaps a hole through the stand	STAND_ROW1_RIGHT	STAND_ROW1_RIGHT_ASSEMBLY
#60	the right Row 1 support that holds the bearing and guides the gear shaft	STAND_ROW1_RIGHT	FASTENED_STAND_ROW1_RIGHT_ASSEMBLY
#61	the white crank-side gear stand with both a bearing slot and a penetrating hole	STAND_ROW1_RIGHT	STAND_ROW1_RIGHT_ASSEMBLY
#63	Row 1 right bracket	STAND_ROW1_RIGHT	STAND_ROW1_RIGHT_ASSEMBLY
#64	right gear stand of Row 1	STAND_ROW1_RIGHT	STAND_ROW1_RIGHT_ASSEMBLY
#82	a fastener used to secure a Row 1 bearing stand to the baseboard	SCREW_ROW1	FASTENED_STAND_ROW1_RIGHT_ASSEMBLY
#88	the Row 2 gear shaft	GEAR_ROD_ROW2	GEAR_ROD_ROW2_ASSEMBLY
#89	the long red cylindrical shaft with two pin holes	GEAR_ROD_ROW2	GEAR_ROD_ROW2_ASSEMBLY
#90	the shaft crossing Row 2 between the two red brackets	GEAR_ROD_ROW2	GEAR_ROD_ROW2_ASSEMBLY
#91	the shaft that passes through both Row 2 bearings and carries the small and medium gears	GEAR_ROD_ROW2	GEAR_ROD_ROW2_ASSEMBLY
#92	the long red triangle-row gear shaft seated in both bearing brackets	GEAR_ROD_ROW2	GEAR_ROD_ROW2_ASSEMBLY
#93	the Row 2 shaft that carries the small and medium gears and transfers rotation between the two stands	GEAR_ROD_ROW2	GEAR_ROD_ROW2_ASSEMBLY
#94	the long red cylindrical rod with two retaining-pin holes spanning the second row from Row 1	GEAR_ROD_ROW2	GEAR_ROD_ROW2_ASSEMBLY
#95	red gear rod	GEAR_ROD_ROW2	GEAR_ROD_ROW2_ASSEMBLY
#96	Row 2 shaft	GEAR_ROD_ROW2	GEAR_ROD_ROW2_ASSEMBLY
#97	gear rod of Row 2	GEAR_ROD_ROW2	GEAR_ROD_ROW2_ASSEMBLY

#	Referring expression	Correct answer	Raw model output
#98	red gear rod in the triangle row	GEAR_ROD_ROW2	GEAR_ROD_ROW2_ASSEMBLY
#104	the red triangle-row left gear stand supporting the gear shaft	STAND_ROW2_LEFT	STAND_ROW2_LEFT_ASSEMBLY
#112	the Row 2 right gear stand	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT_ASSEMBLY
#113	the upright red right stand with a circular bearing opening	STAND_ROW2_RIGHT	STAND_ROW2_RIGHT_ASSEMBLY
#116	the red triangle-row right gear stand supporting the gear shaft	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT_ASSEMBLY
#118	the upright red bearing bracket on the right, beside the medium right gear in the triangle-marked row	STAND_ROW2_RIGHT	STAND_ROW2_RIGHT_ASSEMBLY
#121	right gear stand of Row 2	STAND_ROW2_RIGHT	GEAR_ROW2_RIGHT_ASSEMBLY
#149	a fastener used to secure a Row 2 bearing stand to the baseboard	SCREW_ROW2	FASTENED_STAND_ROW2_RIGHT_ASSEMBLY
#155	the Row 3 gear shaft	GEAR_ROD_ROW3	GEAR_ROD_ROW3_ASSEMBLY
#158	the shaft that passes through both Row 3 bearings and carries the small and medium gears	GEAR_ROD_ROW3	GEAR_ROD_ROW3_ASSEMBLY
#160	the Row 3 shaft that carries the small and medium gears and transfers rotation between the two stands	GEAR_ROD_ROW3	GEAR_ROD_ROW3_ASSEMBLY
#161	the long green cylindrical rod with two retaining-pin holes spanning Row 3	GEAR_ROD_ROW3	GEAR_ROD_ROW3_ASSEMBLY
#164	gear rod of Row 3	GEAR_ROD_ROW3	GEAR_ROD_ROW3_ASSEMBLY
#165	green gear rod in the hexagon row	GEAR_ROD_ROW3	GEAR_ROD_ROW3_ASSEMBLY
#167	the Row 3 left gear stand	STAND_ROW3_LEFT	STAND_ROW3_LEFT_ASSEMBLY
#168	the upright green left stand with a circular bearing opening	STAND_ROW3_LEFT	STAND_ROW3_LEFT_ASSEMBLY
#170	the Row 3 support bracket where the left bearing fits, next to the small left gear	STAND_ROW3_LEFT	STAND_ROW3_LEFT_ASSEMBLY
#172	the left Row 3 support that holds a bearing and supports the gear shaft	STAND_ROW3_LEFT	STAND_ROW3_LEFT_ASSEMBLY
#179	the Row 3 right gear stand	STAND_ROW3_RIGHT	STAND_ROW3_RIGHT_ASSEMBLY
#180	the upright green right stand with a circular bearing opening	STAND_ROW3_RIGHT	STAND_ROW3_RIGHT_ASSEMBLY
#181	the right bearing support bracket in Row 3	STAND_ROW3_RIGHT	STAND_ROW3_RIGHT_ASSEMBLY
#183	the green hexagon-row right gear stand supporting the gear shaft	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT_ASSEMBLY
#185	the upright green bearing bracket on the right, beside the medium right gear in the hexagon-marked row	STAND_ROW3_RIGHT	STAND_ROW3_RIGHT_ASSEMBLY
#186	green right stand	STAND_ROW3_RIGHT	STAND_ROW3_RIGHT_ASSEMBLY
#187	Row 3 right bracket	STAND_ROW3_RIGHT	STAND_ROW3_RIGHT_ASSEMBLY
#188	right gear stand of Row 3	STAND_ROW3_RIGHT	STAND_ROW3_RIGHT_ASSEMBLY
#189	green right gear stand in the hexagon row	STAND_ROW3_RIGHT	GEAR_ROW3_RIGHT_ASSEMBLY

#	Referring expression	Correct answer	Raw model output
#216	a fastener used to secure a Row 3 bearing stand to the baseboard	SCREW_ROW3	FASTENED_STAND_ROW3_RIGHT_ASSEMBLY
#224	the shaft crossing Row 4 between the two blue brackets	GEAR_ROD_ROW4	GEAR_ROD_ROW4_ASSEMBLY
#225	the shaft that passes through both Row 4 bearings and carries one large gear	GEAR_ROD_ROW4	GEAR_ROD_ROW4_ASSEMBLY
#227	the Row 4 shaft that carries one large gear and transfers rotation between the two stands	GEAR_ROD_ROW4	GEAR_ROD_ROW4_ASSEMBLY
#230	Row 4 shaft	GEAR_ROD_ROW4	GEAR_ROD_ROW4_ASSEMBLY
#234	the Row 4 left gear stand	STAND_ROW4_LEFT	STAND_ROW4_LEFT_ASSEMBLY
#235	the upright blue left stand with a circular bearing opening	STAND_ROW4_LEFT	STAND_ROW4_LEFT_ASSEMBLY
#236	the left bearing support bracket in Row 4	STAND_ROW4_LEFT	STAND_ROW4_LEFT_ASSEMBLY
#237	the Row 4 support bracket where the left bearing fits, next to the large left gear	STAND_ROW4_LEFT	STAND_ROW4_LEFT_ASSEMBLY
#238	the blue square-row left gear stand supporting the gear shaft	STAND_ROW4_LEFT	STAND_ROW4_LEFT_ASSEMBLY
#239	the left Row 4 support that holds a bearing and supports the gear shaft	STAND_ROW4_LEFT	STAND_ROW4_LEFT_ASSEMBLY
#240	the upright blue bearing bracket on the left, beside the large left gear in the square-marked row	STAND_ROW4_LEFT	STAND_ROW4_LEFT_ASSEMBLY
#242	Row 4 left bracket	STAND_ROW4_LEFT	STAND_ROW4_LEFT_ASSEMBLY
#243	left gear stand of Row 4	STAND_ROW4_LEFT	STAND_ROW4_LEFT_ASSEMBLY
#244	blue left gear stand in the square row	STAND_ROW4_LEFT	STAND_ROW4_LEFT_ASSEMBLY
#246	the Row 4 right gear stand	STAND_ROW4_RIGHT	STAND_ROW4_RIGHT_ASSEMBLY
#247	the upright blue right stand with a circular bearing opening	STAND_ROW4_RIGHT	STAND_ROW4_RIGHT_ASSEMBLY
#248	the right bearing support bracket in Row 4	STAND_ROW4_RIGHT	STAND_ROW4_RIGHT_ASSEMBLY
#249	the Row 4 right stand where the bearing fits, at the right end of the blue gear rod	STAND_ROW4_RIGHT	STAND_ROW4_RIGHT_ASSEMBLY
#250	the blue square-row right gear stand supporting the gear shaft	STAND_ROW4_RIGHT	STAND_ROW4_RIGHT_ASSEMBLY
#251	the right Row 4 support that holds a bearing and supports the gear shaft	STAND_ROW4_RIGHT	FASTENED_STAND_ROW4_RIGHT_ASSEMBLY
#252	the upright blue bearing bracket on the right, beside the crank-side end of the shaft in the square-marked row	STAND_ROW4_RIGHT	STAND_ROW4_RIGHT_ASSEMBLY
#253	blue right stand	STAND_ROW4_RIGHT	STAND_ROW4_RIGHT_ASSEMBLY
#254	Row 4 right bracket	STAND_ROW4_RIGHT	STAND_ROW4_RIGHT_ASSEMBLY
#255	right gear stand of Row 4	STAND_ROW4_RIGHT	STAND_ROW4_RIGHT_ASSEMBLY
#256	blue right gear stand in the square row	STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT_ASSEMBLY
#257	right bracket of the blue square row	STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT

#	Referring expression	Correct answer	Raw model output
#273	a fastener used to secure a Row 4 bearing stand to the baseboard	SCREW_ROW4	FASTENED_STAND_ROW4_RIGHT_ASSEMBLY

3. Row/side over-specification 49 — answered a specific identifier (e.g. BEARING_ROW2_LEFT) instead of the grouped label (BEARING/PIN/SCREW_ROW{n})

#	Referring expression	Correct answer	Raw model output
#11	the radial shaft-support bearing	BEARING	BEARING_ROW1_RIGHT_ASSEMBLY
#12	the small black circular ring with a center hole	BEARING	BEARING_ROW2_LEFT
#13	the round black part centered inside a bearing support bracket	BEARING	BEARING_ROW2_LEFT
#14	the bearing through which a gear shaft passes	BEARING	BEARING_ROW1_RIGHT_ASSEMBLY
#15	the small black ring-shaped bearing that fits into each colored stand	BEARING	BEARING_ROW2_LEFT
#16	the black bearing that supports a rotating gear shaft inside a stand	BEARING	BEARING_ROW2_RIGHT
#17	the small black ring seated in the round opening of a colored support bracket	BEARING	BEARING_ROW2_LEFT
#18	black bearing	BEARING	BEARING_ROW2_RIGHT
#19	circular bearing	BEARING	BEARING_ROW1_RIGHT_ASSEMBLY
#79	a Row 1 H5 hex-socket mounting screw	SCREW_ROW1	SCREW_ROW1_RIGHT
#80	a small silver screw with a recessed H5 hex-socket head	SCREW_ROW1	SCREW_ROW1_RIGHT
#81	a screw at the base of a Row 1 gear stand	SCREW_ROW1	SCREW_ROW1_RIGHT
#83	the silver H5 hex-socket screw for the white Row 1 stand	SCREW_ROW1	SCREW_ROW1_RIGHT
#84	the Row 1 fastener tightened with a 5 mm hex key	SCREW_ROW1	SCREW_ROW1_RIGHT
#85	the small silver screw located at the foot of a white support bracket	SCREW_ROW1	SCREW_ROW1_RIGHT
#86	Row 1 screw	SCREW_ROW1	SCREW_ROW1_RIGHT
#87	H5 screw	SCREW_ROW1	SCREW_ROW1_RIGHT
#146	a Row 2 T25 Torx mounting screw	SCREW_ROW2	SCREW_ROW2_RIGHT
#147	a small black screw with a recessed T25 Torx head	SCREW_ROW2	SCREW_ROW2_RIGHT
#148	a screw at the base of a Row 2 gear stand	SCREW_ROW2	SCREW_ROW2_RIGHT
#150	the black T25 Torx screw for the red Row 2 stand	SCREW_ROW2	SCREW_ROW2_RIGHT
#151	the Row 2 fastener tightened with a T25 driver	SCREW_ROW2	SCREW_ROW2_RIGHT
#152	the small black screw located at the foot of a red support bracket	SCREW_ROW2	SCREW_ROW2_RIGHT
#153	Row 2 screw	SCREW_ROW2	SCREW_ROW2_RIGHT
#154	T25 screw	SCREW_ROW2	SCREW_ROW2_RIGHT

#	Referring expression	Correct answer	Raw model output
#213	a Row 3 H3 hex-socket mounting screw	SCREW_ROW3	SCREW_ROW3_RIGHT
#214	a small black screw with a recessed H3 hex-socket head	SCREW_ROW3	SCREW_ROW3_RIGHT
#215	a screw at the base of a Row 3 gear stand	SCREW_ROW3	SCREW_ROW3_RIGHT
#217	the black H3 hex-socket screw for the green Row 3 stand	SCREW_ROW3	SCREW_ROW3_RIGHT
#218	the Row 3 fastener tightened with a 3 mm hex key	SCREW_ROW3	SCREW_ROW3_RIGHT
#219	the small black screw located at the foot of a green support bracket	SCREW_ROW3	SCREW_ROW3_RIGHT
#220	Row 3 screw	SCREW_ROW3	SCREW_ROW3_RIGHT
#221	H3 screw	SCREW_ROW3	SCREW_ROW3_RIGHT
#270	a Row 4 Phillips mounting screw	SCREW_ROW4	SCREW_ROW4_RIGHT
#272	a screw at the base of a Row 4 gear stand	SCREW_ROW4	SCREW_ROW4_LEFT
#274	the silver Phillips screw for the blue Row 4 stand	SCREW_ROW4	SCREW_ROW4_RIGHT
#275	the Row 4 fastener tightened with a Phillips screwdriver	SCREW_ROW4	SCREW_ROW4_RIGHT
#276	the small silver screw located at the foot of a blue support bracket	SCREW_ROW4	SCREW_ROW4_RIGHT
#277	Row 4 screw	SCREW_ROW4	SCREW_ROW4_RIGHT
#279	the wooden retaining dowel	PIN	PIN_ROW1_RIGHT
#280	the short thin brown cylindrical peg	PIN	PIN_ROW1_RIGHT
#281	the small peg positioned crosswise through a gear-shaft hole	PIN	PIN_ROW2_LEFT
#282	the dowel that prevents a gear or crank from sliding along its shaft	PIN	PIN_ROW1_RIGHT
#283	the small brown cylindrical retaining pin fitted beside a gear	PIN	PIN_ROW2_LEFT
#284	the wooden retainer inserted through a shaft hole to stop an attached part sliding	PIN	PIN_ROW1_RIGHT
#285	the short brown dowel crossing the gear rod beside a gear or the crank	PIN	PIN_ROW1_RIGHT
#286	wooden pin	PIN	PIN_ROW1_RIGHT
#287	brown pin	PIN	PIN_ROW4_LEFT
#288	retaining pin	PIN	PIN_ROW1_RIGHT

4. Correct, but blocked by reasoning chain 0 — the model identified the right part, but the raw output still contains a <think> block or other wrapping, so strict grading (which compares the full raw text) can never match

None.